Course notes on behavioural economics

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Preface

In this online book, I examine the foundations of behavioural economics.

The book is based on an undergraduate subject I teach in as part of UTS's Bachelor of Economics program, 23005 Behavioural Economics.

There are no mathematical prerequisites for this subject. Hence, the mathematics is kept at a basic level.

This book takes a traditional approach to behavioural economics, providing the basic economics foundations and examining how modifications to the core economic axioms lead to new insight. There is ample critique of the field (as you can read here), but this book is not primarily about that critique. Those foundations can be the launchpad for a critique, and without understanding those foundations the critique can be hollow.

This book covers the following areas:

- **Economic foundations**: In this part I introduce some core economic concepts that we will use and modify in exploring behavioural economics
- **Decision making under risk and uncertainty**: In this part I introduce the traditional economic approach to decision making under risk and uncertainty, and provide some empirical anomalies that arise when using this approach.
- **Prospect theory**: In this part I describe the pre-emininent alternative to expected utility theory, prospect theory. I also provide examples and possible applications.
- **Intertemporal choice**: In this part I explore decision making involving costs and benefits occurring at different times. I look at two types of time preference: exponential discounting and quasi-hyperbolic discounting.

What is behavioural economics?

Behavioural economics is a discipline that seeks to increase the explanatory power of traditional economic approaches by incorporating more realistic psychological foundations.

Behavioural economics retains the general framework and tools used by economists, such as utility maximisation, but deviates from some assumptions to generate new insights and better predictions. These deviations are generally grounded in observed behaviour rather than abstract principles.

As behavioural economics revisits our fundamental understanding of economic behaviour, it has implications for all areas of economics.

What is not behavioural economics?

The term "behavioural economics" is commonly used to describe work that is better described as psychology or "behavioural science".

Economics is the study of how economic agents make decisions under conditions of scarcity and of the interactions between those agents.

Behavioural science is the study of human behaviour. Behavioural economics is not just about studying human behaviour, but about how a better understanding of human behaviour could improve the study of the decisions of economic agents and their interactions.

Similarly, psychology is the study of the human mind, decisions and behaviour. It is part of the behavioural sciences. Psychology is broad with many discrete fields. Behavioural economics tends to draw on a small subset of psychological research in refining the economic view on human behaviour.

Three thought experiments

Many early behavioural economics experiments were built on thought experiments concerning human behaviour that was hard to explain with traditional economic frameworks.

Below are three thought experiments to illustrate the types of problems that concern behavioural economists. In this book I will examine potential explanations for each of these behaviours.

Thought experiment 1

This first thought experiment comes from Kahneman and Tversky (1984).

A. Imagine that you have decided to see a play and paid the admission price of $100 per ticket. As you enter the theatre, you discover that you have lost the ticket. The seat was not marked, and the ticket cannot be recovered.

Would you pay $100 for another ticket?

B. Imagine that you have decided to see a play where admission is $100 per ticket. As you enter the theatre, you discover that you have lost a $100 bill.

Would you still pay $100 for a ticket for the play?

The two situations appear equivalent, yet people are more likely to pay $100 for a ticket in scenario B.

Thought experiment 2

This second thought experiment comes from Thaler (1980).

Mr. R bought a case of good wine … for about $5 a bottle. A few years later his wine merchant offered to buy the wine back for $100 a bottle. He refused, although he has never paid more than $35 for a bottle of wine.

Why is there such a large difference between the price at which he is willing to buy and the price at which he is willing to sell?

Thought experiment 3

This third thought experiment also comes from Thaler (1980).

A group of hungry economists is awaiting dinner when a large can of cashews is opened and placed on the coffee table. After half the can is devoured in three minutes, everyone agrees to put the rest of the cashews into the pantry.

Why did they agree to remove the cashews when they simply could have not eaten them? Why did they deliberately reduce their choice set?

Part I

Economic foundations

As behavioural economics builds on the frameworks and tools used by economists, in this part I introduce some of the core economic concepts that we will build on or explicitly deviate from in our exploration of behavioural economics.

I will:

- Examine the preference relation used by economists
- Discuss the foundational axioms of rationality
- Describe how the preference relation and axioms form the basis of utility functions

1 Preferences

Economists uses the preference relation to capture the ordering that an agent gives to alternatives between which they might choose:

- Strong (or strict) preference ($\succ$): "is better than"
- Weak preference ($\succcurlyeq$): "is at least as good as"
- Indifference ($\sim$): "is as good as" or "the agent is indifferent between". $x \sim y$ if and only if $x \succcurlyeq y$ and $y \succcurlyeq x$.

I might have the following preference:

- banana $\succ$ apples

I strongly prefer bananas to apples.

Similarly:

- bananas $\sim$ oranges

I am indifferent between bananas and oranges.

2 Rationality

A standard economic assumption is that decision makers are rational.

However, rationality in economics has a different definition to the 'lay' definition of rationality.

Rationality simply means that preferences respect some elementary principles (axioms) of desirable behaviour. Economists tend to keep the number of assumptions as small as possible.

The two major assumptions are completeness and transitivity.

2.1 Completeness

Completeness means that a consumer can always compare any two options. The consumer cannot fail to have a preference between two options (although that preference may be indifference).

For example, if a consumer was presented with a choice between an apple and a banana, or a choice between a Mercedes and a BMW, they will always strictly prefer one of them or be indifferent between the two. They will never not know what to choose or be unable to make a choice. They cannot be indecisive.

Formally, we can state the completeness axiom as follows:

> For all x, y, either $x \succcurlyeq y$ or $y \succcurlyeq x$ (or both).

Completeness means people always have a preference for x or y, or are indifferent between the two.

While the completeness axiom appears sensible, it is possible to develop examples where it may not hold. Consider a choice between two possible holiday destinations. Or two potential dates or spouses. If you are torn between the options and unable to make up your mind, this would represent a breach of the completeness axiom.

Incomplete preferences are different from indifference. If you are indifferent, you would be able to quickly decide by, say, flipping a coin. You will be equally satisfied whatever the outcome. Incompleteness makes choice impossible.

2.2 Transitivity

Under transitivity, if a person prefers A to B and B to C, they will prefer A to C.

Formally, we can state the transitivity axiom as follows:

> For all x, y, z, if $x \geq y$ and $y \geq z$, then $x \geq z$.

One classic argument for transitivity is the concept of a "money pump" (Davidson et al. (1955)). Suppose you have a person who prefers A to B, B to C and C to A (i.e. intransitive preferences). They have \$20 and an endowment of C. They are offered B in exchange for their endowment of C for some small nominal cost (say \$1). If they make the trade they now have B and \$19. They are then offered A in exchange for their endowment of B, again for a nominal cost. Finally, they are offered C in exchange for their endowment of A for a further nominal cost. They now have an endowment of C and \$17. This process can be repeated until the agent has no money.

2.3 Preference Orderings

If we assume preferences are rational (transitive and complete), it is possible to construct a preference ordering.

Completeness guarantees that there will be only one ordering.

Transitivity guarantees that there will be no cycles in strict preference. Weak preferences can cycle, so that one can prefer a to b and b to c and c to a, but this entails indifference.

3 Utility functions

Economists often use numbers to represent strength of preference. This is done through utility functions.

A utility function associates a number with each member of the universe. For example:

- Banana: 3
- Orange: 2
- Apple: 1

This does not mean that I rate bananas three times higher than apples. This utility scale is ordinal, not cardinal. The following is equivalent:

- Banana: 300
- Orange: 2
- Apple: 1

Formally, the utility function $u(\cdot)$:

- maps the set of alternatives into the set of real numbers
- assigns larger numbers to preferred alternatives.

That is:

$x \succcurlyeq y \iff u(x) \geq u(y)$

$x \succ y \iff u(x) > u(y)$

$x \sim y \iff u(x) = u(y)$

Under the axioms of completeness and transitivity, the consumer behaves *as if* they have a utility function $u(x_i)$ over outcomes x_i.

4 Economic foundation exercises

4.1 Question 1.

The paradox of Buridan's ass runs as follows: An ass that is equally hungry and thirsty is placed halfway between a pile of hay and a bucket of water. The ass cannot decide between the hay and water, so dies of dehydration and starvation.

What axioms of choice are relevant to this fable? Are any axioms violated?

Question 1 answer

The axiom of completeness is violated. Under this axiom an agent cannot fail to have a preference between two options (although that preference may be indifference). Incompleteness is different from indifference. Is the ass was merely indifferent it would be happy taking either option and be equally satisfied. Indifference does not make a choice impossible.

4.2 Question 2

You are considering two mobile phone plans. Each has different monthly fees, data caps, excess data charges, international inclusions and 5G coverage. You realise it will take all day to work through the fine print to undserstand the plans.

You decide that your options are:

a) Pick a plan by flipping a coin.

b) Spend the day working through the plans and choose one.

c) Avoid the work by reading a book instead.

Does this decision accord with the axioms we have discussed?

Question 2 answer

All three choices could be considered to accord with the axioms we have discussed. What we have effectively done in each case is created a richer choice set. In no case are their preferences incomplete. You could think of the choice set as {Plan A, Plan B, invest to understand plans, do something else}.
You would only state that the preferences are incomplete if the agent wasn't able to express a preference between Plan A and Plan B. However, if they were forced to choose and happy to flip a coin, that would suggest indifference.

4.3 Question 3

Consider the following statements about rationality in economics. How do these criticisms relate to the definition of rationality we have discussed?

a) From Robert Frank in the New York Times:

> TRADITIONAL economic models assume that people are self-interested in the narrow sense. If "homo economicus" - the stereotypical rational actor in these models - finds a wallet on the sidewalk, he keeps the cash inside. He doesn't leave tips after dining in restaurants that he will never visit again. And he would never vote in a presidential election, much less make an anonymous donation of money or time to a presidential campaign.

Question 3a) answer

The traditional economic axioms assume self-interest in a narrow sense, in that people make decisions in accordance with their preferences. However, completeness and transitivity say nothing about the content of those preferences. A person might prefer to return the wallet or leave a large tip for the good feeling they get. They might enjoy voting and care about outcomes for others.
Even auxiliary axioms such as monotonicity or non-satiation leave these possibilities open in that while the agent will always want more, they do not require that it is purely for their own benefit.

b) From Brian Easton in interest.co.nz:

> For the last 150 years much economic analysis has been based on homo economicus, an 'economic' man who is rational and narrowly self-interested and who pursues his subjectively defined ends optimally.

Question 3b) answer

First, "man who is rational" accords with the technical definition in economics but differs from how used in common speech (and likely that of readers of the article).
If we read "narrowly self-interested" to mean that they make decisions in accordance with their preferences, then we might agree with that statement. However, we cannot place any further content into that idea of self-interest.
Similarly, the statement "pursues his subjectively defined ends optimally" accords with the idea that under the axioms of completeness and transitivity, the consumer behaves as if they have a utility function $U(x_i)$ over outcomes x_i. They are able to choose between any two options. They also do not make errors (although randomness can be built into utility functions). This definition of "optimally" is narrower than might be used in common speech.

Part II

Decision making under risk and uncertainty

In this part I examine the basic economic approach to decision making under risk and uncertainty.

I introduce the basic axioms on which the economic approach is based, discuss expected utility theory, and provide a set of empirical anomalies in expected utility theory that provide grounds for the behavioural economic approach.

Notation

At times when discussing decision making under risk and uncertainty, I will use the following notation:

Suppose we have a lottery L that gives outcomes $x_1, x_2, ..., x_n$ with probabilities $p_1, p_2, ..., p_n$. We can write this as:

$$L = (x_1, p_1; x_2, p_2; ...; x_n, p_n)$$

For example:

$$L = (-100, 0.5; 200, 0.5)$$

means that the gambler loses 100 with 50% probability and wins 200 with 50% probability.

5 Expected value

The expected value of a gamble is the amount you can expect to win on average, in the long run, when you play a gamble.

Example: We flip a coin. I give you \$1 if it is heads and you give me \$1 if it is tails. What is the expected value of this gamble?

The expected value is \$0. You lose \$1 half the time and gain \$1 half the time.

Formally, given a gamble, the expected value $E[X]$ of the random variable X is:

$$\begin{aligned} E[X] &= p_1 x_1 + p_2 x_2 + ... + p_n x_n \\ &= \sum_{i=1}^{n} p_i x_i \end{aligned}$$

p_i is the probability of x_i.

5.1 Expected value examples

5.1.1 Example 1

You are offered bet with a 50% chance of winning \$10 and a 50% chance of losing \$8.

The expected value of the gamble is:

$$\begin{aligned} E[X] &= \sum_{i=1}^{n} p_i x_i \\ &= 0.5(10) + 0.5(-8) \\ &= \$1 \end{aligned}$$

Your chance of winning increases to 60%. The expected value of the gamble is:

$$
\begin{aligned}
E[X] &= \sum_{i=1}^{n} p_i x_i \\
&= 0.6(10) + 0.4(-8) \\
&= \$2.80
\end{aligned}
$$

5.1.2 Example 2

You are offered a bet with a 50% bet chance of winning 50% of your wealth and a 50% of chance of losing 40% of your wealth.

The expected value of the bet is:

$$
\begin{aligned}
E[X] &= \sum_{i=1}^{n} p_i x_i \\
&= 0.5 * (0.5W) + 0.5 * (-0.4W) \\
&= 0.05W
\end{aligned}
$$

The expected value is 5% of your wealth.

6 Axioms for Expected Utility Theory

For decision making under risk, we require two additional axioms of desirable behaviour to develop a predictive or descriptive model. These additional axioms are continuity and the independence of irrelevant alternatives.

Under the axioms of:

- Completeness
- Transitivity
- Continuity
- Independence

a decision maker behaves as if choosing between risky prospects by selecting the one with the highest expected utility.

The four axioms of completeness, transitivity, continuity and independence are often called the von Neumann–Morgenstern axioms for a rational decision maker. You can see now we have a second potential benchmark of "rationality".

Note: Preferences under these assumptions are cardinal.

Beyond those four axioms, some additional axioms are often adopted for practical purposes. These are also discussed below.

6.1 Continuity

The idea behind continuity is that people have similar preferences for similar bundles. If x is preferred to y, bundles close to x are preferred to bundles close to y. There are no "jumps" in utility.

Continuity guarantees that every preference relation can be represented by a continuous utility function (and vice versa).

Here are two formal definitions.

6.1.1 Definition 1

1. A preference relation is continuous if for any $x \succ y$ there exists a number $\epsilon > 0$ such that every bundle a that is less distant from x than ϵ and every bundle b that is less distant from y than ϵ results in $a \succ b$.

To put this another way, a preference relation is continuous if for any $x \succ y$ there are *some* neighbourhoods $N_\epsilon x$ and $N_\epsilon y$ around x and y such that for every $a \in N_\epsilon x$ and $b \in N_\epsilon x$ we have $a \succ b$.

One way to picture this is to imagine a circle around bundle x of radius ϵ. This circle represents the neighbourhood. There will always exist some circle - even if very small - within which every bundle a is preferred to bundle y.

The intuition behind this definition is that a very small change in your bundle should not result in a sudden switch of your preferences. If you prefer 5 bananas to 2 oranges, you will likely prefer 4.9 bananas to 2 oranges. (And if not, there will be some amount of bananas between 4.9 and 5 that you prefer over 2 oranges.)

Here's another intuitive example: if you prefer a Mercedes to a Toyota, there will be *some* level of defect in the Mercedes that you would be willing to accept while still preferring the Mercedes to the Toyota.

6.1.2 Definition 2

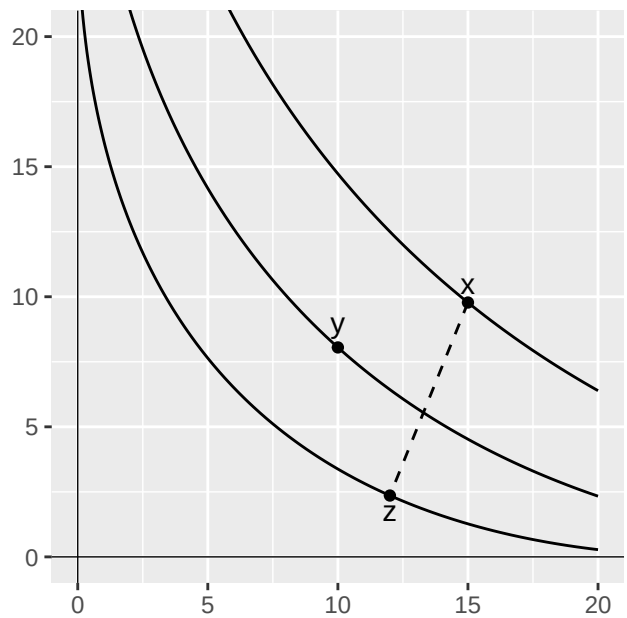
If x, y and z are lotteries with $x \succcurlyeq y \succcurlyeq z$, the continuity axiom requires that there exists a probability p such that y is equally as good as a mix of x and z. That is, there exists p such that:

$$px + (1-p)z \sim y$$

The below diagram illustrates continuity under this definition.

On the diagram are three bundles: x, y and z, and each sits on a different indifference curve. The indifference curve that x is on is higher than that of y which is higher than that of z. That is, $x \succcurlyeq y \succcurlyeq z$.

Now consider a gamble that pays x with probability p and z with probability $1-p$. Each value of p would result in a gamble with utility falling between that of x and z. If we were to draw a line between x and z, you could think of the utility of the gamble for each value of p as having the same utility as a bundle on that line. Under the continuity axiom, there would be no holes in that line. For some value of p, that gamble will be on the same indifference curve for y. At that point, $px + (1-p)z \sim y$.



Note: If you read the literature about expected utility, you may come across an axiom called the Archimedean property. Only one of continuity or the Archimedean property need be assumed. I will not cover the Archimedean property in this book.

6.1.3 Example of discontinuous preferences

Lexiographic preferences occur where an agent prefers any amount of a good x to any amount of another good y. If choosing between bundles of goods, the agent will choose the bundle with the most x, regardless of the amount of y. They will only consider the amount of y if the amount of x in two bundles is identical.

Consider an agent with lexiographic preferences who is offered the following combinations of x and y.

A. (1, 1)

B. (1, 2)

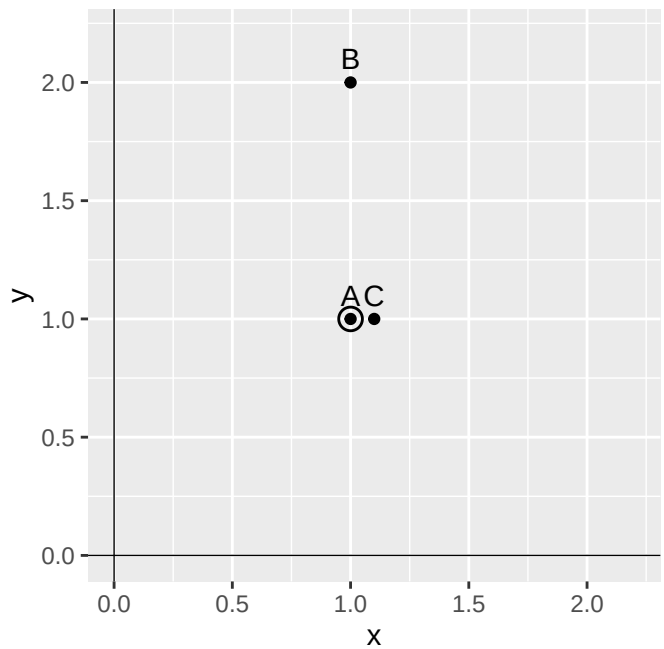
C. (1.1, 1)

Their preference ranking will be: $C \succ B \succ A$.

This function is not continuous as there is a "jump" whenever there is an increase in x, even if y is large. Add an infinitesimal amount ϵ of x to bundle A and the preference relation between bundle A and bundle B flips.

These three bundles A, B and C are represented graphically below.

First, let's consider these preferences in terms of Definition 1. Around A I have drawn a circle of radius ϵ (the neighbourhood). No matter how small I draw this circle (no matter how small ϵ), any bundle within the circle that lies to the right of A (that is, contains $x > 1$) is preferred to bundle B. There is a jump in preferences to the right of A.



To consider this in terms of the second formal definition, there is no p for which:

$$pA + (1-p)C \sim B$$

When $p = 1$, $B \succ A$. For any $p < 1$, $pA + (1-p)C \succ B$ as any non-zero share of C makes the combination of A and C preferred.

One interesting feature of lexiographic preferences is that you cannot draw indifference curves on this figure. If a bundle differs from another, it must be strictly preferred to the other as no amount of y can make up for any amount of x.

6.2 Independence of irrelevant alternatives

Under independence of irrelevant alternatives, a person who mixes two gambles with an irrelevant third gamble will maintain the same order of preference between those two original gambles as when the two are presented independently of the third.

Formally, if:

- x and y are lotteries with $x \succcurlyeq y$ and
- p is the probability that a third option z is present, then:

$$pz + (1-p)x \succcurlyeq pz + (1-p)y$$

The third choice, z, is irrelevant. The order of preference for x before y holds. It is independent of the presence of z.

6.2.1 Example of independence of irrelevant alternatives

Suppose x is an orange and y is an apple. I strictly prefer an orange to an apple.

Suppose there is not a third possibility z of receiving no fruit. Under the independence of irrelevant alternatives, if I prefer oranges to apples, I will prefer a gamble with a 50% chance of getting an orange and 50% chance of receiving nothing to a gamble with a 50% chance of getting an apple and a 50% chance of receiving nothing.

That is:

$$\text{orange} \succ \text{apple} \implies 50\% \text{ chance of orange} \succ 50\% \text{ chance of apple}$$

6.3 Auxilliary axioms

Beyond completeness, transitivity, continuity and independence, economists often adopt other axioms. These are not required for expected utility theory, but make analysis more practicable.

These include:

- Utility is calculated over total wealth.
- Non-satiation: No matter what you have, there is always another (nearby) bundle that you would rather have.
- Monotonicity: Preferences are monotone if more of any good in the bundle makes the agent strictly better off. Non-satiation is implied by monotonicity, but not the other way around. Monotonicity implies downward sloping indifference curves.
- Convexity: Convexity means that people have a preference for variety or combination (indifference curves bulge toward the origin).

- Diminishing marginal utility: Each successive additional unit of consumption delivers a smaller (diminishing) amount of utility than the last.

I provide further detail on convexity and diminishing marginal utility below.

6.3.1 Convexity

Convexity means that people have a preference for variety or combination (indifference curves bulge toward the origin). Averages are better than extremes.

In many contexts this makes sense. For example, suppose an agent is indifferent between two beers and two meat pies. Under this assumption, any mix of the two (e.g. a beer and a pie) will be at least as preferred as the two beers or two pies.

Formally, a preference relation is convex if, for any $x \succcurlyeq y$ and for every $\theta \in [0, 1]$:

$$\theta x + (1 - \theta)y \succcurlyeq y$$

Strict convexity is where, for any $x \sim y$, $x \neq y$ and for every $\theta \in [0, 1]$:

$$\theta x + (1 - \theta)y \succcurlyeq y$$

$$\theta x + (1 - \theta)y \succcurlyeq x$$

For two equivalent goods or bundles, a weighted average of the two bundles is better than each of those bundles.

6.3.2 Diminishing marginal utility

Marginal utility is how much utility you get or lose from an incremental decrease or increase in consumption. Under diminishing marginal utility, each successive additional unit of consumption delivers a smaller (diminishing) amount of utility than the last.

Diminishing marginal utility leads to risk averse preferences.

Diminishing marginal utility is related to the axiom of convexity. Diminishing marginal utility will lead to convex indifference curves. However, the reverse relationship does not always hold.

7 Expected Utility Theory

7.1 Maximising expected utility

People do not seek to maximize expected value, but instead maximize expected utility.

Under expected utility theory, people choose between risky prospects (another word for lottery or gamble common in the literature) by comparing expected utility values. The agent maximises expected utility, $E[U(x)]$:

$$\begin{aligned} E[U(x)] &= p_1 U(x_1) + p_2 U(x_2) + ... + p_n U(x_n) \\ &= \sum_{i=1}^{n} p_i U(x_i) \end{aligned}$$

p_i is the probability of outcome x_i.

$U(x_i)$ is the utility of outcome x_i

Under Expected Utility Theory, risky options are evaluated in 3 steps:

1. Define utility over final outcomes $x_1, ..., x_n$
2. Weight utility of each outcome $U(x_i)$ by the probability of outcome p_i
3. Add the weighted utilities

There is an important note to make here regarding outcomes $x_1, ..., x_n$. Typically, these outcomes are not just the payoffs from the gamble, but rather the agent's final position. If the agent has wealth of \$100 and is offered a coin flip to win or lose \$10, the outcomes are typically taken to be \$90 and \$110. Their decision depends on their current wealth. As a result, expected utility is often represented as:

$$\begin{aligned} E[U(W+x)] &= p_1 U(W+x_1) + p_2 U(W+x_2) + ... + p_n U(W+x_n) \\ &= \sum_{i=1}^{n} p_i U(W+x_i) \end{aligned}$$

7.2 Attitudes toward risk

Expected utility theory captures an individual's risk preference by means of their utility function.

If a person prefers a sure amount over a gamble with comparable expected value, they are risk averse.

If a person prefers the gamble over the sure amount, they are risk seeking.

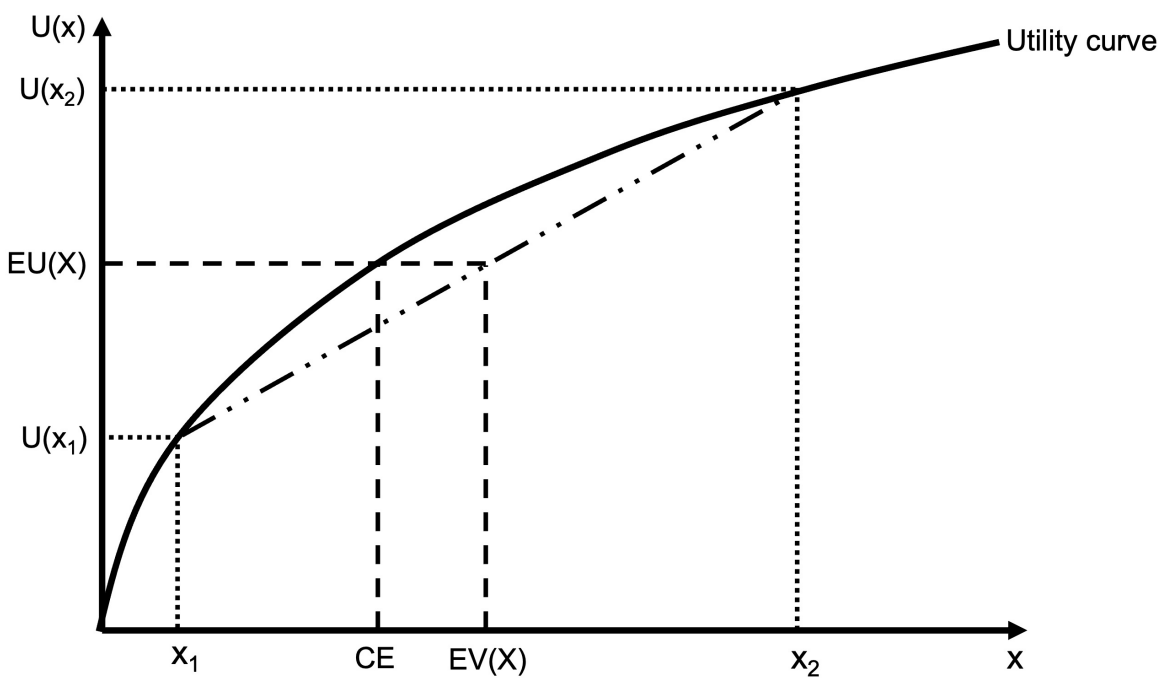
If a person is indifferent, they are risk neutral.

The certainty equivalent (CE) of a gamble X is the the amount of money such that you are indifferent between taking the gamble and taking the money:

$$u(CE) = EU(X)$$

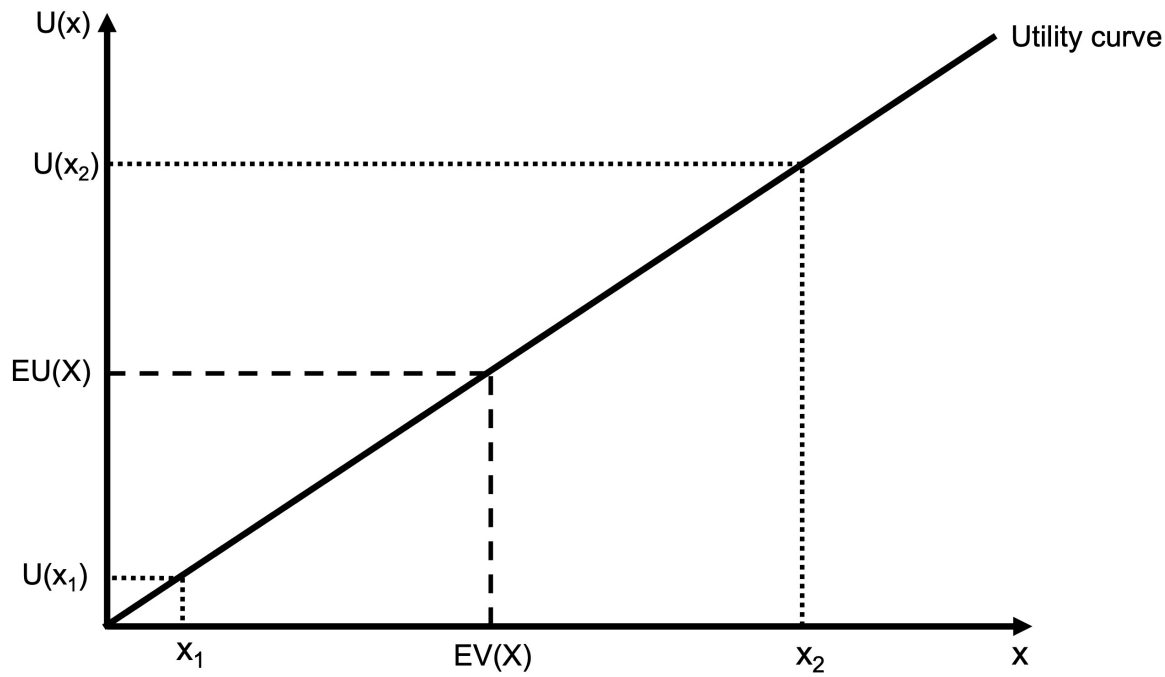
7.2.1 Risk aversion

For a risk averse person, the certainty equivalent is less than the expected value of the gamble.



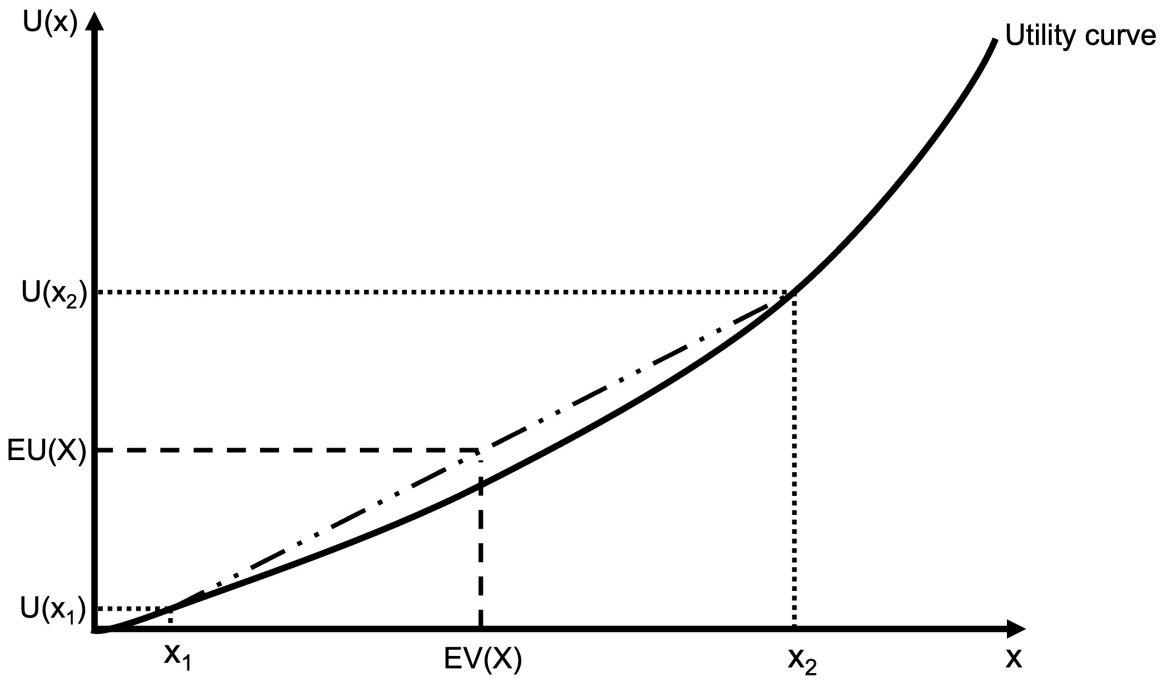
7.2.2 Risk neutrality

For a risk averse person, the certainty equivalent is equal to the expected value of the gamble. A risk neutral person is an expected value maximiser.



7.2.3 Risk seeking

For a risk seeking person, the certainty equivalent is more than the expected value of the gamble. The gamble has value in and of itself.


U(x)
Utility curve
U(x2)
EU(X)
U(x1)
x1
EV(X)
x2
x

8 Subjective Expected Utility

Subjective expected utility theory is used in situations of uncertainty, where the probabilities are not known. People maximize expected utility using a subjective estimate of probability.

The agent maximises subjective expected utility, $E[U(x)]$:

$$\begin{aligned} E[U(x)] &= \pi(x_1)U(x_1) + \pi(x_2)U(x_2) + ... + \pi(x_n)U(x_n) \\ &= \sum_{i=1}^{n} \pi(x_i)U(x_i) \end{aligned}$$

where $\pi(x_i)$ is the subjective probability of x_i.

People have different beliefs about the probabilities of different outcomes, but these typically come from a "sound" foundation such as Bayesian probability theory. Decision makers are Bayesian, in that, given the evidence, they have beliefs which are consistent with Bayes' rule.

In Subjective Expected Utility Theory, uncertain options are evaluated in 4 steps:

1. Define utility over final outcomes $x_1, ..., x_n$.
2. Define subjective probability $\pi(x_i)$ over final outcomes $x_1, ..., x_n$.
3. Weight utility of each outcome $U(x_i)$ by the subjective probability $\pi(x_i)$.
4. Add the weighted utilities.

9 Expected utility examples

9.1 Example 1

Suppose your utility function is $U(x) = ln(x)$.

You have a 50% chance of winning \$10 and a 50% chance of losing \$10. Assume your starting wealth is \$20.

What is the expected value of this game?

$$\begin{aligned} E[x] &= \sum_{i=1}^{n} p_i x_i \\ &= 0.5 * 10 + 0.5 * (-10) \\ &= 0 \end{aligned}$$

The expected value of the game is \$0.

What is the expected utility of this game?

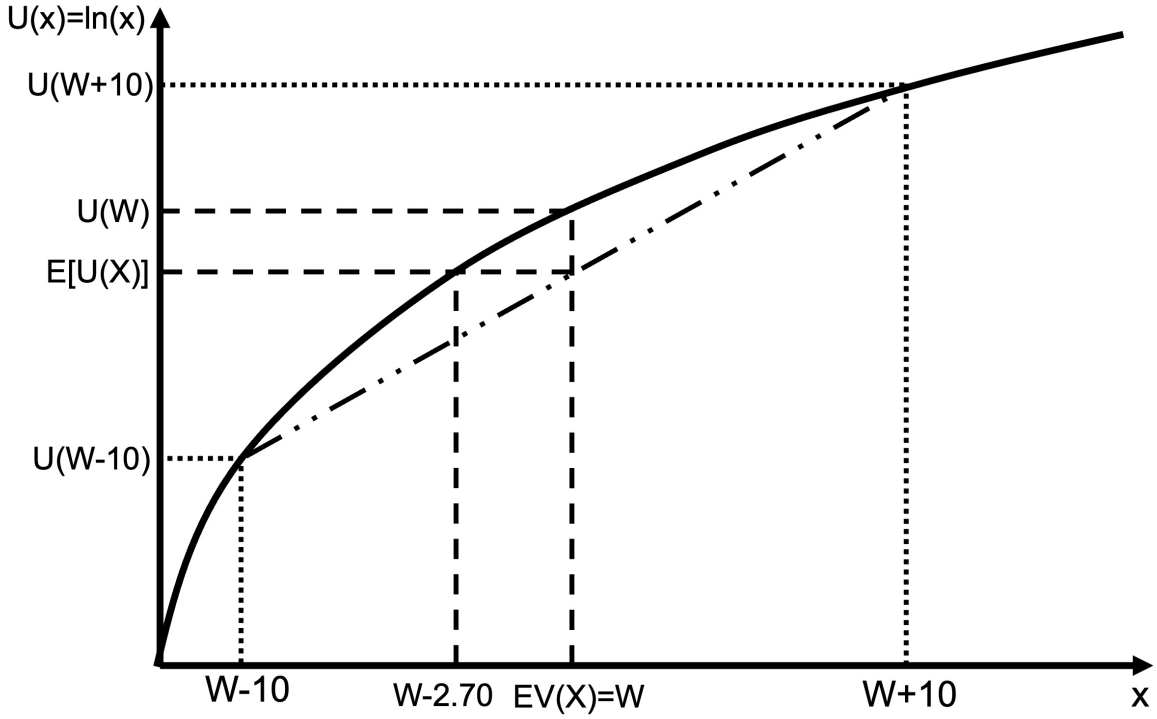
$$\begin{aligned} E[U(W+x)] &= \sum_{i=1}^{n} p_i U(x_i + W) \\ &= 0.5U(20-10) + 0.5U(20+10) \\ &= 0.5ln(10) + 0.5ln(30) \\ &= 2.85 \end{aligned}$$

What does an expected utility of 2.85 mean? To make it tangible, we can ask what wealth would give that utility.

$$U(W) = ln(W) = 2.85$$

$$W = e^{2.85} = \$17.30$$

This gamble with expected value of zero reduces utility by an amount equivalent to \$2.70.



We could also say that the certainty equivalent of this gamble is final wealth of \$17.30, a loss of \$2.70.

9.2 Example 2

Suppose your utility function is $U(x) = ln(x)$.

You have an 80% chance of winning \$10 and a 50% chance of losing \$10. Assume your starting wealth is \$20.

What is the expected value and the expected utility of this game?

$$
\begin{aligned}
E[x] &= \sum_{i=1}^{n} p_i x_i \\
&= 0.8 * 10 + 0.2 * (-10) \\
&= \$6
\end{aligned}
$$

The expected value of the game is \$6.

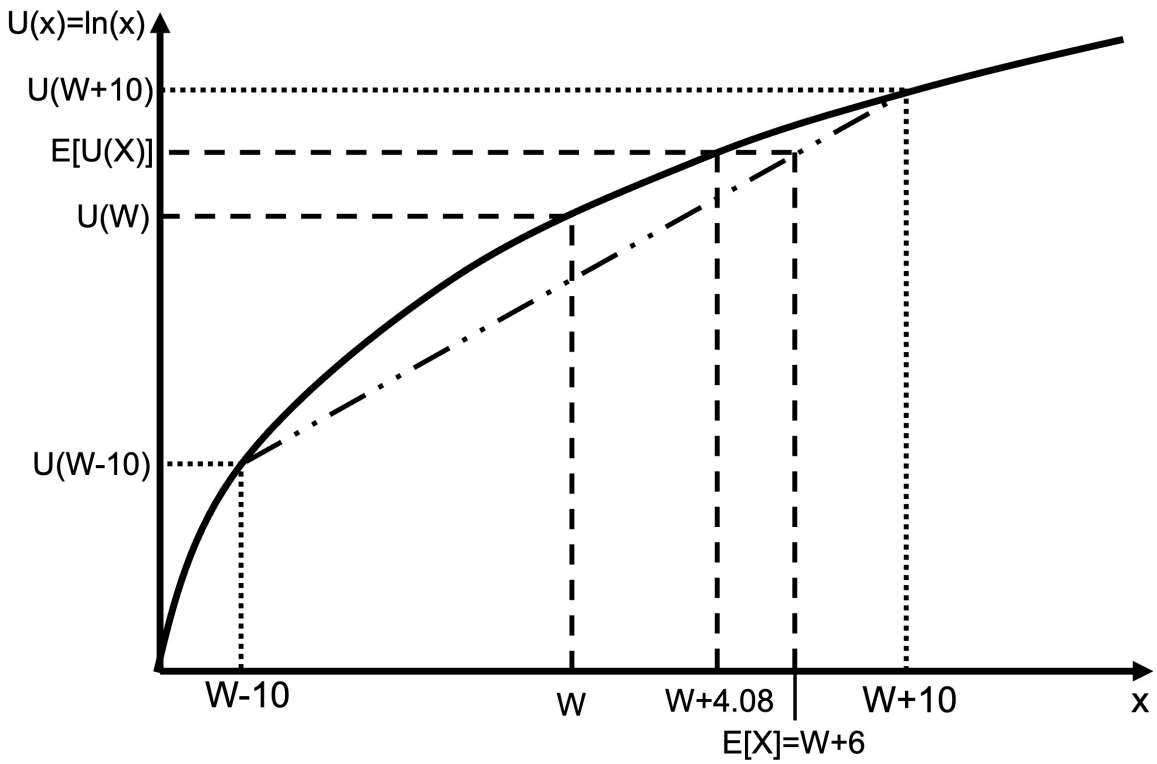
$$
\begin{aligned}
E[U(W+x)] &= \sum_{i=1}^{n} p_i U(x_i + W) \\
&= 0.8U(20+10) + 0.2U(20-10) \\
&= 0.8ln(30) + 0.2ln(10) \\
&= 3.18
\end{aligned}
$$

What does an expected utility of 3.18 mean? To make it tangible, we can ask what wealth would give that utility.

$$U(W) = ln(W) = 3.18$$

$$W = e^{3.18} = \$24.08$$

This gamble with expected value of \$6 increases utility by an amount equivalent to \$4.08.



9.3 Example 3

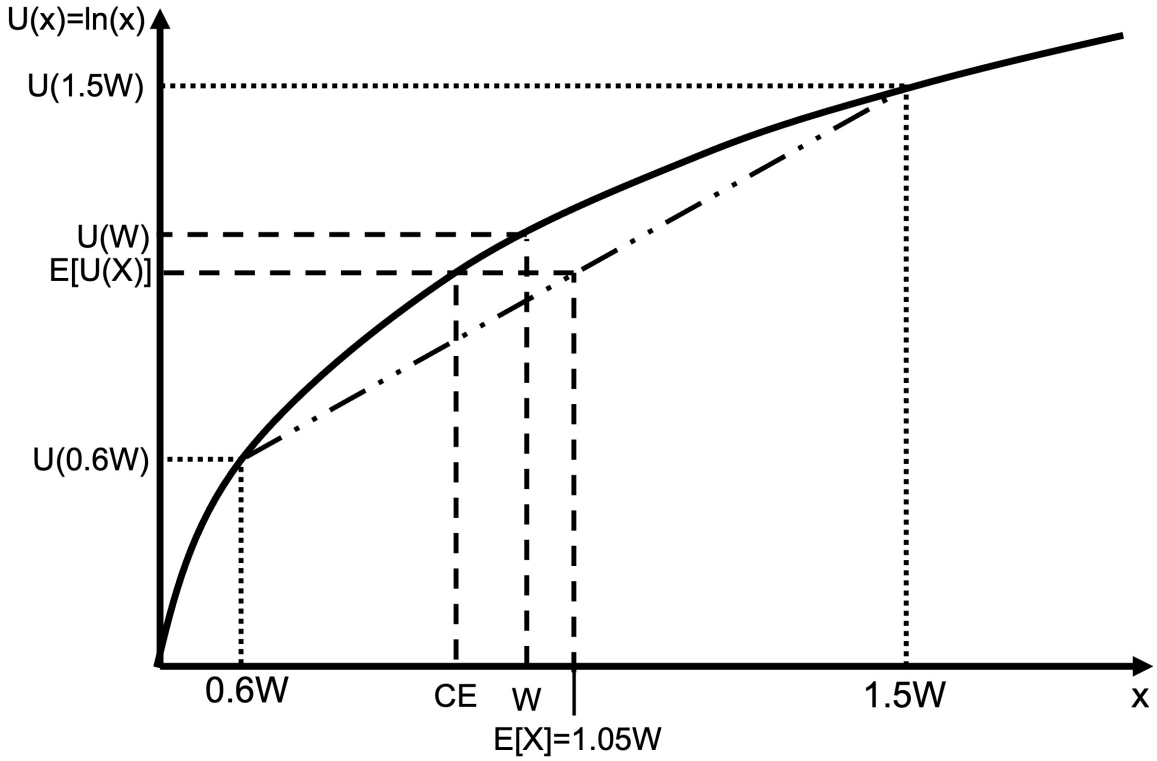
Suppose your utility function is $U(x) = ln(x)$.

You have a 50% chance of increasing your wealth by 50% and a 50% chance of decreasing your wealth by 40%.

What is the expected value and the expected utility of this games?

$$
\begin{aligned}
E[U(X)] &= \sum_{i=1}^{n} p_i U(X_i) \\
&= 0.5U(0.6W) + 0.5U(1.5W) \\
&= 0.5ln(0.6) + 0.5 * ln(W) + 0.5ln(1.5) + 0.5 * ln(W) \\
&= -0.255 + 0.203 + ln(W) \\
&= -0.053 + ln(W)
\end{aligned}
$$

Here we have a gamble with a positive expected value, 5% of your wealth, but lower expected utility. Someone with log utility would reject this bet.



9.4 Example 4: The St. Petersburg game

The St. Petersburg game was invented by Swiss mathematician Nicolas Bernoulli.

The game starts with a pot containing \$2. A dealer then flips a coin. The pot doubles every time a head appears. The game ends and the player win the pot as soon as tails appears.

- Tail on the first flip: \$2
- Tail on the second flip: \$4
- Tail on the third flip: \$8
- Tail on the fourth flip: \$16

And so on.

The expected value of this game is equal to:

$$E[X] = \underbrace{\frac{1}{2} * 2}_{\text{Tail first}} + \underbrace{(\frac{1}{2} * \frac{1}{2}) * 4}_{\text{Tail second}} + \underbrace{(\frac{1}{2} * \frac{1}{2} * \frac{1}{2}) * 8}_{\text{Tail third}}$$

$$+ \underbrace{(\frac{1}{2} * \frac{1}{2} * \frac{1}{2} * \frac{1}{2}) * 16}_{\text{Tail fourth}} + ...$$

$$= 1 + 1 + 1 + 1 + ...$$
$$= \sum_{k=1}^{\infty} 1$$
$$= \infty$$

The $\sum$ operator means "sum for $k = 1$ to $k = \infty$".

The expected utility of this game is equal to:

$$E[U(X)] = \underbrace{\frac{1}{2} * U(W+2)}_{\text{Tail first}} + \underbrace{(\frac{1}{2} * \frac{1}{2}) * U(W+4)}_{\text{Tail second}}$$

$$+ \underbrace{(\frac{1}{2} * \frac{1}{2} * \frac{1}{2}) * U(W+8)}_{\text{Tail third}} + \underbrace{(\frac{1}{2} * \frac{1}{2} * \frac{1}{2} * \frac{1}{2}) * U(W+16)}_{\text{Tail fourth}} + ...$$

$$= \frac{1}{2}U(W+2) + \frac{1}{4}U(W+4) + \frac{1}{8}U(W+8) + \frac{1}{16}U(W+16) + ...$$

$$= \sum_{k=1}^{k=\infty} \frac{1}{2^k} U(W + 2^k)$$

What is the maximum $\$c$ a risk neutral player with $U(x) = x$ would be willing to pay to play the game? One strategy to determine $\$c$ is to ask at what $\$c$ the player would be indifferent between accepting and rejecting a chance to play. That is the maximum $\$c$ that they would be willing to pay. They will be indifferent when $U(W) = E[U(X - c)]$.

$$U(W) = E[U(X-c)]$$

$$U(W) = \sum_{k=1}^{k=\infty} \frac{1}{2^k}(U(W + \$2^k - c)$$

$$W = \sum_{k=1}^{k=\infty} \frac{1}{2^k}(W + 2^k - c)$$

$$W = W - c + \sum_{k=1}^{k=\infty} 1 \qquad \left(\text{as } \sum_{k=1}^{k=\infty} \frac{1}{2^k} = 1\right)$$

$$c = \infty$$

A risk-neutral player would pay any amount $\$c$ to play.

What is the maximum $\$c$ a risk averse player with $U(x) = ln(x)$ player would be willing to pay to play the game? How does their wealth affect their willingness to pay?

Again we will determine at what $\$c$ the player is indifferent between accepting and rejecting a chance to play, which occurs when $U(W) = E[U(X-c)]$.

$$U(W) = E[U(X-c)]$$

$$U(W) = \sum_{k=1}^{k=\infty} \frac{1}{2^k} U(W + \$2^k - c)$$

$$ln(W) = \sum_{k=1}^{k=\infty} \frac{1}{2^k} ln(W + \$2^k - c)$$

There is no closed form solution to this equation to enable us to determine c. It needs to be solved via numerical methods (such as testing and iterating to a solution). Someone who has wealth of \$0.01 would be willing to pay up to \$2.01 (they would need to borrow). Someone with wealth \$1000 would be willing to pay \$10.95, while a person with wealth of \$1 million would be willing to pay \$20.87.

Note: we cannot solve for a person with no wealth as ln(0) is undefined.

Another way to gain an intuition of what is happening to ask what is the utility of a risk averse player whose only asset is the opportunity to play this game.

$$
\begin{aligned}
E[U(X)] &= \sum_{k=1}^{k=\infty} \frac{1}{2^k} U(\$2^k) \\
&= \sum_{k=1}^{k=\infty} \frac{1}{2^k} ln(2^k) \\
&= \sum_{k=1}^{k=\infty} \frac{k}{2^k} ln(2) \\
&= \frac{1}{2} ln(2) + \frac{2}{4} ln(2) + \frac{3}{8} ln(2) + \frac{4}{16} ln(2) + \frac{5}{32} ln(2) + ... \\
&= (\frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{1}{4} + \frac{5}{32} + ...) ln(2) \\
&= 2ln(2)
\end{aligned}
$$

The change in utility from each flip rapidly converges as the series of fractions sums to two. The expected utility from the game is equal to the utility of \$4 ($U(W) = ln(W) = 2ln(2); W = e^{2ln2} = 4$).

Why does willingness to pay increase with wealth?

With log utility, as wealth increases the slope of the log function increasingly approximates a linear function (the second derivative approaches zero). Hence, the gambler displays less risk averse (closer to risk neutral) behaviour.

10 Anomalies of expected utility theory

10.1 The Allais Paradox

Choice 1: Choose one of the following bets:

Bet A:

- \$2500 with probability: 33%
- \$2400 with probability: 66%
- \$0 with probability: 1%

Bet B:

- \$2400 with probability: 100%

Choice 2: Choose one of the following bets:

Bet C:

- \$2500 with probability: 33%
- \$0 with probability: 67%

Bet D:

- \$2400 with probability: 34%
- \$0 with probability: 66%

When Kahneman and Tversky (1979) ran this experiment 82% of participants chose option B and 83% of participants chose option C.

According to Expected Utility Theory, if an agent selects B:

$$U(2400) > 0.33U(2500) + 0.66U(2400) + 0.01U(0)$$

$$0.34U(2400) > 0.33U(2500) + 0.01U(0)$$

According to Expected Utility Theory, if an agent selects C:

$$0.33U(2500) + 0.67U(0) > 0.34U(2400) + 0.66U(0)$$

$$0.33U(2500) + 0.01U(0) > 0.34U(2400)$$

This is a contradiction. Under expected utility theory, if an agent chooses A it should choose C. And if the agent chooses B it should choose D. This phenomena is called the Allais Paradox after Maurice Allais (1953), who first identified it.

Why does this occur? What axiom is being breached?

Here is another representation of the choices.

Choice 1				Choice 2			
A		B		C		D	
Payoff	Chance	Payoff	Chance	Payoff	Chance	Payoff	Chance
$2400	66%	$2400	100%	$0	67%	$0	66%
$0	1%					$2400	34%
$2500	33%			$2500	33%		

This second tabular representation is equivalent. I have split the outcomes in options B and C.

Choice 1				Choice 2			
A		B		C		D	
Payoff	Chance	Payoff	Chance	Payoff	Chance	Payoff	Chance
$2400	66%	$2400	66%	$0	66%	$0	66%
$0	1%	$2400	34%	$0	1%	$2400	34%
$2500	33%			$2500	33%		

You can see that the bets contained in the bottom two rows are the same. For choice 1, they are paired with a 66% chance of winning $2400, and for choice 2 a 66% chance of winning $0.

Choice 1				Choice 2			
A		B		C		D	
Payoff	Chance	Payoff	Chance	Payoff	Chance	Payoff	Chance
$2400	66%	$2400	66%	$0	66%	$0	66%
$0	1%	$2400	34%	$0	1%	$2400	34%
$2500	33%			$2500	33%		

Preferring B to A and C to D is a violation of the axiom of the independence of irrelevant alternatives: Under that axiom, two gambles mixed with an irrelevant third gamble will maintain the same order of preference as when the two are presented independently of the third gamble.

Recall that the formal definition for the independence of irrelevant alternatives axiom is that if:

- x and y are lotteries with $x \succcurlyeq y$ and
- p is the probability that a third option z is present, then:

$$pz + (1-p)x \succcurlyeq pz + (1-p)y$$

For each of the choices in our lottery:

- $p = 66$
- x is a 100% chance of \$2400
- y is a 0.01/(1−0.66) chance of \$0 and 0.33/(1−0.66) chance of \$2500
- z is \$2400 in choice 1 and \$0 in choice 2, although z's value should not matter due to its assumed irrelevance.

10.2 Absurd rates of risk aversion

I offer you the following one-off bet by flipping a coin:

Head: You win \$550

Tail: You lose \$500

Would you accept this bet?

Barberis et al. (2006) offered a real \$550:\$500 bet to experimental participants including those with substantial wealth (MBA students, hedge fund managers, etc.).

70% of the sample turned down the bet, included professional investors with wealth above \$10 million.

Under the axiom of diminishing marginal utility, we would conclude people are risk averse to small bets.

But, for sufficiently high levels of wealth, $U(x)$ is approximately linear and people always take favourable bets.

The minimum $U(x)$ curvature required to reconcile an investor with \$10 million investor downing a 50:50 bet as small as +\$550 or -\$500 would imply that agents reject immensely favourable bets, which is not realistic.

Rabin (2000) showed that rejection of bets over moderate stakes can require absurd rates of risk aversion. For instance, if a person who acts consistent with expected utility theory always

turns down a 50:50 bet to win $110 or lose $100 whatever their initial level of wealth, they will also turn down a 50:50 bet to win $1 billion, lose $1,000.

10.3 Framing

Kahneman and Tversky (1984) reported the following experiment.

> A group of experimental participants were shown the following:
>
> Imagine that the U.S. is preparing for the outbreak of an unusual Asian disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Assume that the exact scientific estimates of the consequences of the programs are as follows:
>
> If Program A is adopted, 200 people will be saved.
>
> If Program B is adopted, there is a one-third probability that 600 people will be saved and a two-thirds probability that no people will be saved.
>
> Which of the two programs would you favour?

72% of participants chose option A.

Another group of experimental participants were shown the following:

> Imagine that the U.S. is preparing for the outbreak of an unusual Asian disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Assume that the exact scientific estimates of the consequences of the programs are as follows:
>
> If Program C is adopted, 400 people will die.
>
> If Program D is adopted, there is a one-third probability that nobody will die and a two-thirds probability that 600 people will die.
>
> Which of the two programs would you favour?

22% of participants chose option C.

72% of participants chose A and 22% of participants chose option C. Yet these two are equivalent.

10.4 Reference points

Consider the following two scenarios:

- You have not checked your share portfolio in a while. You expect it is worth around \$40,000. Today when you check, it is worth \$30,000. Do you feel rich or poor?
- You have not checked your share portfolio in a while. You expect it is worth around \$20,000. Today when you check, it is worth \$30,000. Do you feel rich or poor?

Under expected utility theory, those two scenarios should feel the same as you have $U(\$30{,}000)$ in both cases.

11 Risk and uncertainty exercises

11.1 Question 1: Expected value

You are playing roulette at the casino. There are 37 numbered pockets around the edge of the wheel (0 through 36). If you make a straight up bet on one of the 37 single numbers, you are paid \$35 for every dollar you bet. What is the expected value of a \$20 bet.

Question 1 answer

The expected value of the Roulette bet is:

$$\begin{aligned} E[X] &= \sum_{i=1}^{n} p_i x_i \\ &= \frac{36}{37} * (-\$20) + \frac{1}{37} * (35 * \$20) \\ &= \$ - 0.54 \end{aligned}$$

11.2 Question 2: Expected value

An agent is considering insurance against bushfire for its \$1,000,000 house. The house has a 1 in 1000 chance of burning down. An insurer is willing to offer full coverage for \$1100.

a) What is the expected value of purchasing insurance?

b) What is the expected value of not purchasing insurance?

Question 2 answer

a) What is the expected value of purchasing insurance?

$$E[\text{purchase}] = -\text{premium} = -\$1,100$$

The expected value of purchasing insurance is the guaranteed loss of the premium.

b) What is the expected value of not purchasing insurance?

$$\begin{aligned} E[\text{don't}] &= P_{\text{burn}} * -value_{\text{house}} \\ &= -0.001 * 1000000 \\ &= -\$1000 \end{aligned}$$

11.3 Question 3: Expected utility

Anika is an expected utility maximiser with the following utility function:

$$U(x) = \sqrt{x}$$

Anika is offered the following choice:

A) A 50% chance of winning \$10 and a 50% chance of winning nothing
B) \$4 for certain

Anika has zero wealth besides this offer.

a) What is the expected value of option A)?

Question 3(a) answer

The expected value of option A) is:

$$\begin{aligned} E[A] &= \sum_{i=1}^{n} p_i x_i \\ &= 0.5 * \$10 + 0.5 * 0 \\ &= \$5 \end{aligned}$$

b) Will Anika choose A) or B)? Why?

Question 3(b) answer

We need to determine the expected utility of each option. Anika will selection the option with the highest expected utility.
The expected utility of option A) is:

$$
\begin{aligned}
EU(A) &= p_1 U(x_1) + p_2 U(x_2) \\
&= 0.5 * \sqrt{10} + 0.5 * \sqrt{0} \\
&= 1.58
\end{aligned}
$$

The expected utility of option B) is:

$$
\begin{aligned}
EU(B) &= U(4) \\
&= \sqrt{4} \\
&= 2
\end{aligned}
$$

Anika will choose option B) as it gives her higher expected utility. Anika is risk averse.

c) What is the certainty equivalent of option A?

Question 3(c) answer

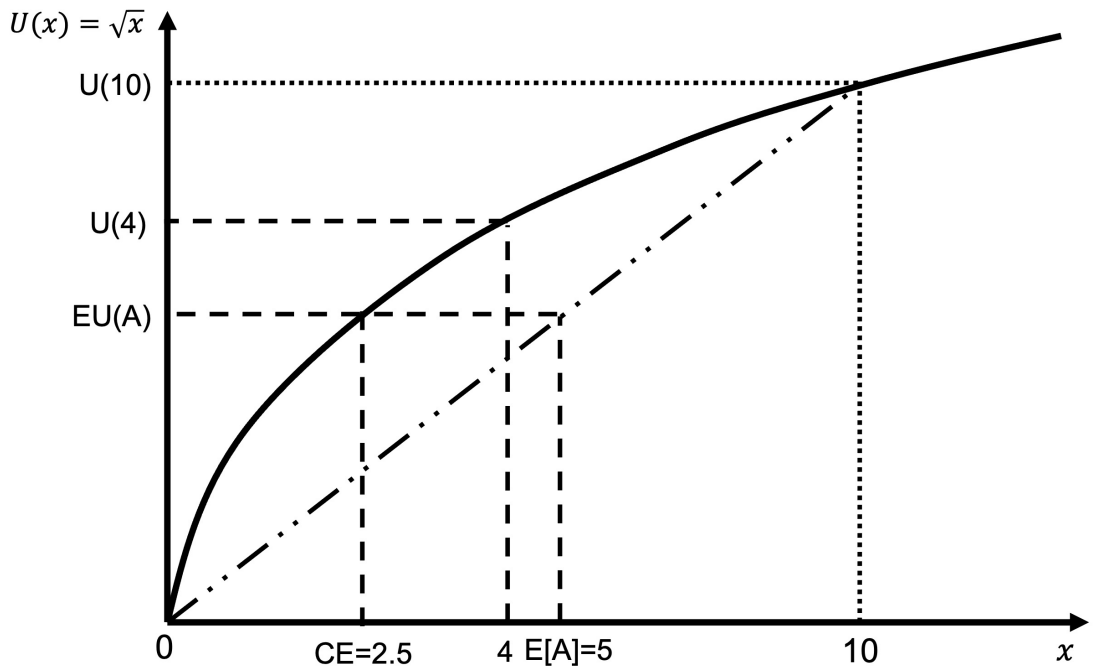
To calculate the certainty equivalent of option A, we calculate what payment with certainty would deliver equivalent expected utility. That is:

$$
\begin{aligned}
EU(CE) &= 1.58 \\
\sqrt{CE} &= 1.58 \\
CE &= 1.58^2 \\
&= 2.5
\end{aligned}
$$

The certainty equivalent of option A is \$2.50. That is, Anika would be indifferent between option A and a payment of \$2.50 for certain.

d) Draw a graph showing Anika's utility curve, the expected value of option A, the expected utility of options A) and B) and the certainty equivalent of option A).

Question 3(d) answer



11.4 Question 4: Expected utility

Consider the following gamble:

(0.5; \$550; 0.5, -\$500)

This gamble provides a 50% chance of winning \$550 and a 50% chance of losing \$500.

a) Would a risk neutral agent (who maximises expected value) be willing to pay \$20 to play this gamble? What is the most they would be willing to pay to play?

Question 4(a) answer

The expected value of the gamble is:

$$
\begin{aligned}
E[X] &= \sum_{i=1}^{n} p_i x_i \\
&= 0.5(550) + 0.5(-500) \\
&= 25
\end{aligned}
$$

This is greater than \$20, so a risk neutral agent will be willing to pay \$20 to participate in the gamble.
We could also have solved this by determining the expected value if they had paid \$20:

$$
\begin{aligned}
E[X] - c &= \sum_{i=1}^{n} p_i x_i - c \\
&= 0.5(550) + 0.5(-500) - 20 \\
&= 5
\end{aligned}
$$

As the expected value is positive, the agent would be willing to pay \$20.

b) Would a risk averse expected utility maximiser with wealth \$1000 and utility function $U(x) = x^{1/2}$ be willing to pay \$20 to play this gamble? What is the most they would be willing to pay to play?

Question 4(b) answer

The expected utility of the gamble for the risk averse agent if they paid \$20 to play is:

$$
\begin{aligned}
EU(x) &= p_1(W + x_1 - c) + p_2(W + x_2 - c) \\
&= 0.5(1000 + 550 - 20)^{1/2} + 0.5(1000 - 500 - 20)^{1/2} \\
&= 30.51
\end{aligned}
$$

The expected utility of not playing the gamble is:

$$
\begin{aligned}
EU(x) &= (1000)^{1/2} \\
&= 31.62
\end{aligned}
$$

They would not pay \$20 as they would have higher utility if they turned down the gamble.

In fact, they would not pay any positive sum to participate in the gamble. If they were offered the gamble for free, their expected utility would be:

$$\begin{aligned} EU(x) &= 0.5(1000 + 550)^{1/2} + 0.5(1000 - 500)^{1/2} \\ &= 30.86 \end{aligned}$$

This is less than if they simply turned down the gamble. They would be willing to pay to avoid the gamble. How much?
We can determine this by asking what wealth a utility of 30.86 is:

$$\begin{aligned} W^{1/2} &= 30.86 \\ W &= 30.51^2 \\ &= \$952.67 \end{aligned}$$

The certainty equivalent of the gamble is \$952.67. The agent would be willing to pay up to \$47.33 to avoid the gamble.

c) Would the expected utility maximiser with utility function $U(x) = x^{1/2}$ change their decision if they had \$1 million in wealth? Explain.

Question 4(c) answer

If they now have \$1 million in wealth, we simply repeat the calculations above with the new wealth.

$$\begin{aligned} EU(x) &= 0.5(1000000 + 550 - 20)^{1/2} + 0.5(1000000 - 500 - 20)^{1/2} \\ &= 1000.00247 \end{aligned}$$

The expected utility of not playing the gamble is:

$$\begin{aligned} EU(x) &= (1000000)^{1/2} \\ &= 1000 \end{aligned}$$

They would be willing to pay \$20 as they would have higher utility if they accepted the gamble.
What is the most they would be willing to pay? If they were offered the gamble for free, their expected utility would be:

$$
\begin{aligned}
EU(x) &= 0.5(1000000 + 550)^{1/2} + 0.5(1000000 - 500)^{1/2} \\
&= 1000.0125
\end{aligned}
$$

How much would they be willing to pay for this opportunity? We can determine this by asking what wealth a utility of 1000.0125 is:

$$
\begin{aligned}
W &= (1000.0124655)^2 \\
&= \$1000024.93
\end{aligned}
$$

The agent would be willing to pay up to \$24.93 for the gamble. This is close to the expected value of \$25.
Intuitively, as the agent's wealth increases their utility function becomes increasingly linear (second derivative approaches zero) and they become closer to risk neutral.

11.5 Question 5: Expected utility: insurance

An agent is considering insurance against bushfire for its \$1,000,000 house. The house has a 1 in 1000 chance of burning down. An insurer is willing to offer full coverage for \$1100.

a) Would a risk neutral agent purchase the insurance?

Question 5a) answer

We have already calculated that purchasing insurance in this case has a lower expected value than not purchasing the insurance. A risk neutral agent would not purchase the insurance.

b) Suppose an agent has a logarithmic utility function ($U(x) = ln(x)$) and they have \$10,000 in cash in addition to their house, giving them wealth (W) of \$1,010,000. Would this agent purchase the insurance? Are they risk seeking, risk neutral or risk averse?

Question 5b) answer

$$
\begin{aligned}
E[U(\text{purchase})] &= ln(W - premium) \\
&= ln(1,008,900) \\
&= 13.8244
\end{aligned}
$$

$$
\begin{aligned}
E[U(\text{don't})] &= 0.999 * ln(W) + 0.001 * ln(W - value_{\text{house}}) \\
&= 0.999 * ln(1,010,000) + 0.001 * ln(10,000) \\
&= 13.8208
\end{aligned}
$$

The expected utility of purchasing insurance is greater than the expected utility from not purchasing insurance. This agent will purchase insurance. They are risk averse.
What is the intuition for this agent's purchase of insurance? Diminishing marginal utility means that the utility of average wealth is greater than the average utility of wealth (e.g. $U(\$0) + U(\$200) < U(\$100) + U(\$100)$). Therefore, their expected utility is higher when wealth is distributed evenly across the possible states of the world rather than concentrated in one state - or in the case of a disaster, very low in one state. The consumer insures as a way of evenly distributing wealth across all possible states.

11.6 Question 6: An anomaly in expected utility

Consider the following two choices:

Choice 1: Choose one of the following bets:

Bet A:

- \$10,000 with probability: 11%
- \$0 with probability: 89%

Bet B:

- \$50,000 with probability: 10%
- \$0 with probability: 90%

Choice 2: Choose one of the following bets:

Bet A':

- \$10,000 with probability: 100%

Bet B':

- \$50,000 with probability: 10%
- \$10,000 with probability: 89%
- \$0 with probability: 1%

Many people pick B for Choice 1 and A' for Choice 2.

Does this pair of choices conform with Expected Utility Theory? Why?

Question 6 answer

According to Expected Utility Theory, if an agent selects B:

$$0.10U(50,000) + 0.90U(0) > 0.11U(10,000) + 0.89U(0)$$

$$0.10U(50,000) + 0.01U(0) > 0.11U(10,000)$$

According to Expected Utility Theory, if an agent selects A':

$$U(10,000) > 0.10U(50,000) + 0.89U(10,000) + 0.01U(0)$$

$$0.11U(10,000) > 0.89U(50,000) + 0.01U(0)$$

This is a contradiction. Under expected utility theory, if an agent chooses B it should choose B'. And if the agent chooses A it should choose A'.
This occurs due to a breach in the principle of independence.
Here is a representation of the choices.

	89%	10%	1%
A	\$0	\$10,000	\$10,000
B	\$0	\$50,000	\$0
A'	\$10,000	\$10,000	\$10,000
B'	\$10,000	\$50,000	\$0

The bets in the two shaded areas are the same. They are paired with an outcomes of either \$10,000 or \$0. Preferring B to A and A' to B' is a violation of the axiom of the independence of irrelevant alternatives: Under that axiom, two gambles mixed with an irrelevant third gamble will maintain the same order of preference as when the two are presented independently of the third gamble.

Using this representation in the table, here is another way of understanding why this combination of choices is an anomaly. Imagine there are 100 tickets numbered 1 to 100. One ticket will be drawn. If a ticket between 1 and 89 is drawn, you win the prize in the first column. If a ticket between 90 and 99 is drawn, you win the amount in the second. If a 100 is drawn, you win the sum in the third.

Suppose that you know the ticket that is drawn is between 1 and 89. Would you prefer A or B? As you would win \$0 with either choice, you will be indifferent. You will similarly be indifferent between A' and B', winning \$10,00 no matter what.

Suppose instead that a ticket between 90 and 100 is drawn, but you know which. You can see that if you prefer A to B, you should also prefer A' to B'. In each choice you are effectively facing the same bet. Let's assume for the moment that you prefer B and B'.

Finally, suppose you don't know what ticket will be drawn. We have just determined that if you know the ticket is between 1 and 89 you are indifferent between the options, but if between 90 and 100 is drawn you prefer B and B'. You do not prefer A or A' when the ticket range is 1 to 89 or 90 to 100, so you should not prefer A or A' when the ticket number is unknown.

Finally, using the formal definition for the independence of irrelevant alternatives axiom:

- if x and y are lotteries with $x \succcurlyeq y$ and
- p is the probability that a third option z is present, then:

$$pz + (1-p)x \succcurlyeq pz + (1-p)y$$

For each of the choices in our lottery:

- $p = 89\%$
- x is a 100% chance of \$10,000
- y is a 0.01/(1-0.89) chance of \$0 and 0.10/(1-0.89) chance of \$50,000
- z is \$10,000 in choice 1 and \$0 in choice 2, although z's value does not matter due to its assumed irrelevance.

Part III

Prospect theory

Kahneman and Tversky (1979) challenged Expected Utility Theory by introducing a new descriptive theory they called Prospect Theory that addresses the anomalies that we identified in the previous part.

Prospect theory has the following features:

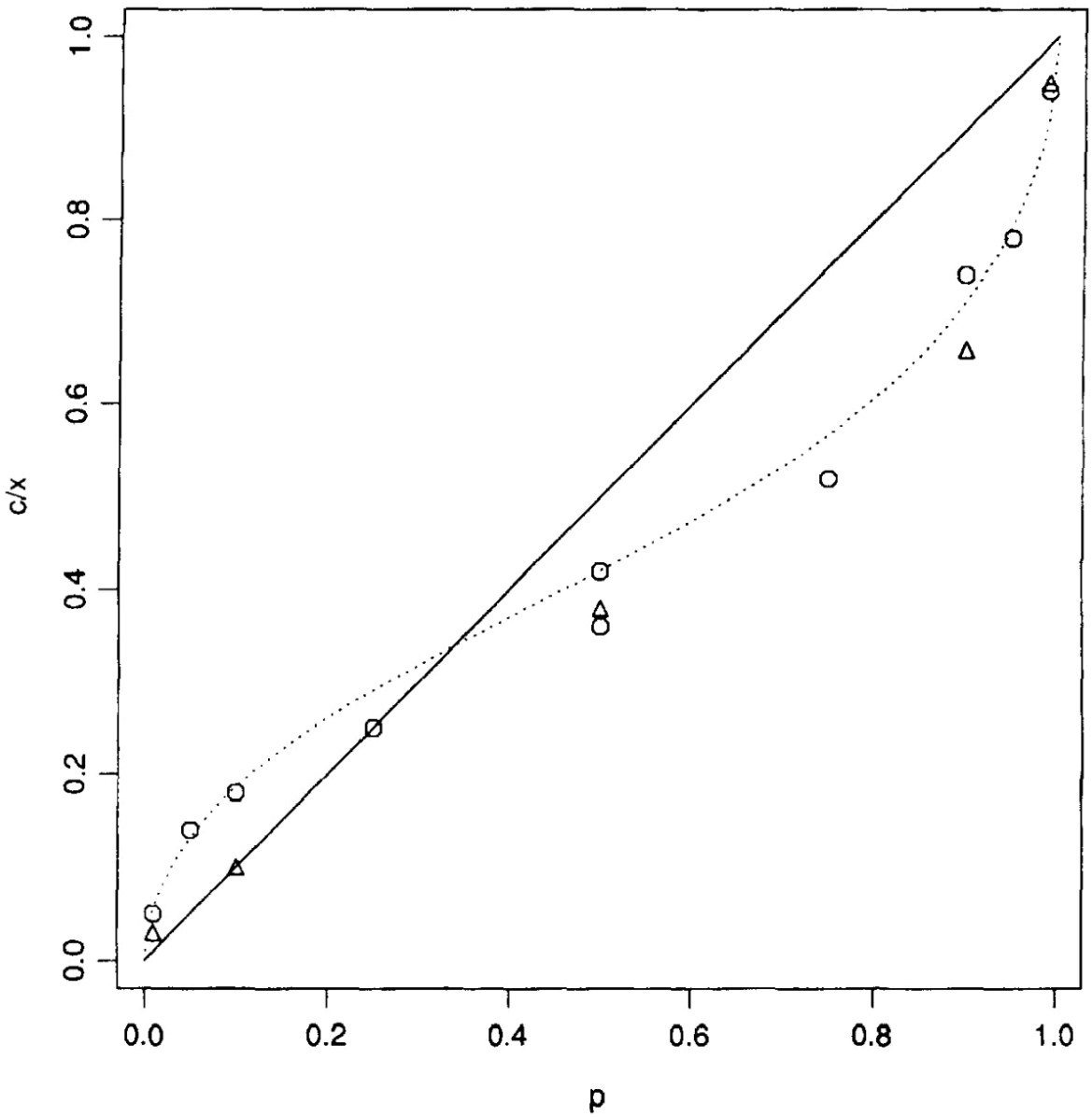
- Probability weighting
- Loss aversion
- The reflection effect
- Reference dependence

I will address each of these features individually before combining them.

12 Probability weighting

In Expected Utility Theory, probabilities enter the expected utility function **linearly**. That is, if an event is twice as likely as another outcome it has double the weight.

Experimental observation confirms that we approximate linear probabilities for intermediate probabilities. But, there is strong evidence that we give overweight certain events (i.e. when $p = 1$) relative to near certain events (e.g. when $p = 0.99$). This is effectively the same as overweighting very low probability events.



Recall our discussion of the Allais Paradox in Section 10.1.

Choice 1: Choose one of the following bets:

Bet A:

- \$2500 with probability: 33%
- \$2400 with probability: 66%
- \$0 with probability: 1%

Bet B:

- \$2400 with probability: 100%

Choice 2: Choose one of the following bets:

Bet C:

- \$2500 with probability: 33%
- \$0 with probability: 67%

Bet D:

- \$2400 with probability: 34%
- \$0 with probability: 66%

If you look at Bet B, this involves a certain outcome, which tennds to be overweighted relative to near-certain events (the 99% chance of \$2400 or \$2500).

Kahneman (2011) calls the large psychological value of the change from 0 to 5% (or some other small probability) the possibility effect. Very unlikely but possible outcomes are given more weight than similar increases in probability for events that are already possible. He calls the large psychological value of the change to 100% the certainty effect. We will pay a lot more for certainty than near certainty.

12.1 The decision weight

Under expected utility theory, the utility of a prospect is a linear expectation. Each possible outcome is weighted by its probability.

In Prospect Theory the decision maker does not weigh probabilities linearly but instead attaches a decision weight $\pi(p_i)$ to each probability.

The new decision-weighting function reflects the empirical regularity we discussed previously: people overweight certain events (i.e. when $p = 1$) relative to near certain events (e.g. when $p = 0.99$) and overweight very low probability events

13 Loss aversion

Loss aversion can instead explain why we behave differently depending on whether the decision making induces a gain or a loss. Loss aversion is the concept that losses loom larger than gains.

People feel more strongly about a loss than they do an equivalent gain, so are often willing to reject gambles with a materially positive expected value.

For instance, if someone feels losses with twice the feeling of gains, a 50:50 bet to win $550, lose $500 will be unattractive. This provides an alternative explanation to the absurd levels of risk aversion required to reject this bet (as discussed in Section 10.2).

13.1 Loss aversion in a deterministic environment: the endowment effect

Kahneman et al. (1991) ran one of the most famous and replicated experiments in economics.

They randomly assigned a free mug to members of a group and asked how much money they would accept for returning the mug (i.e. willingness to accept). The remaining participants were only shown a mug and asked their willingness to pay for the mug.

Kahneman et al. (1991) found that the willingness to accept ($5.75) was substantially higher than the willingness to pay ($2.25).

This experiment has been replicated in real estate markets, the stock market, with basketball tickets and in other domains.

The endowment effect is this phenomena where people impute additional value to the items they own. The endowment effect is argued to be an empirical expression of loss aversion. Willingness to accept is higher as it is payment to incur the loss of the mug.

13.2 Loss aversion in the value function

The following is an example of a value function with reference dependence and loss aversion:

$$v(x) = \begin{cases} x & \text{where} x \geq 0 \\ 2x & \text{where} x < 0 \end{cases}$$

Where x is the outcome relative to the reference point.

For example, suppose someone is given \$100. If their initial reference point is their wealth before receiving the \$100, x will be \$100. Therefore their utility is +100. If the same person instead loses \$100, their utility would be -200.

14 The reflection effect

When people make a risky choice related to gains, they are risk averse. They prefer a certain option with lower expected utility than the expected utility of the risky choice. When making a choice in the loss domain, they become risk seeking. This phenomena is called the reflection effect.

This phenomena might also be thought of as diminishing sensitivity to gains or losses in either direction. This contrasts with expected utility theory where the pain of losses increases as they grow in size.

The reflection effect provides an explanation for the framing effects we observed in the question about the unusual disease. In the gain frame, the preferred option is risk averse. In the loss frame, the preferred option tends to be the risk seeking option.

14.1 The reflection effect in the value function

The following value function is an example of a function where there is diminishing sensitivity to both gains and losses.

$$v(x) = \begin{cases} x^{\frac{1}{2}} \text{where} x \geq 0 \\ -2(-x)^{\frac{1}{2}} \text{where} x < 0 \end{cases}$$

15 Reference dependence

Under Prospect Theory, people assess choices based on a reference point, as opposed to an absolute assessment of their position. Outcomes are coded as losses and gains relative to that reference point.

The earlier example of assessing a share portfolio worth \$30,000 from a reference point of either \$20,000 or \$40,000 is an example of reference dependence.

Under expected utility theory, utility is typically thought of as being the product of your entire wealth (W). For example, if you gain \$100, your utility increases from $u(W)$ to $u(W+\$100)$.

In contrast, under prospect theory, the value function (what the utility function is typically called in Prospect Theory) applies to changes relative to the reference point. If their initial reference point is their wealth before receiving the \$100, the value of that change is $v(\$100)$. The importance of that distinction becomes apparent when we consider how people consider losses and gains.

15.1 Theories of reference depdendence

There are a number of theories on reference point formation. These include:

- The status quo
- Lagged consumption
- Goals
- Recent expectations

15.1.1 The status quo

A common assumption in Prospect Theory is that the reference point is the status quo (as it was for many examples in the original paper by Kahneman and Tversky (1979)). The status quo implies a preference for the current state. Any negative change is perceived as a loss.

The status quo appears straightforward and has been shown to be effective in contexts such as lab experiments.

However, the status quo as a reference point does not appear to be a useful assumption for describing many economic interactions. Suppose you decide to sell your bike. Do you see the foregoing of the bike as a loss?

What is you run a bike shop? Does every sale involve a feeling of loss? For markets in which intangible and fungible goods are exchanged (e.g. stock market) the status quo assumption appears a poor fit.

15.1.2 Lagged consumption

Imagine you win the lottery.

How do you feel one week after the draw?

How do you feel one year after the draw?

Your reference point likely reflects more recent consumption.

Lagged consumption introduces adaptation into reference point determination:

1. We react to shocks
2. The effect of the shock fades in time

15.1.3 Goals

Consider the following problem from Heath et al. (1999):

> Sally and Trish both follow workout plans that usually involve doing 25 sit-ups.
>
> One day, Sally sets a goal of performing 31 sit-ups. She finds herself very tired after performing 35 sit-ups and stops.
>
> Trish sets a goal of performing 39 sit-ups. She finds herself very tired after performing 35 sit-ups and stops.
>
> What emotion is each person experiencing?

With goals as reference points, people see success or failure to achieve a goal as a loss or gain. Although both have the same performance, Sally will have a positive emotional reaction and Trish a negative reaction.

15.1.4 Recent expectations

In this approach, the reference point is beliefs about future outcomes (Kőszegi and Rabin (2006)). For example, a 5% pay rise when expecting a 10% pay rise may be perceived as a loss.

The expectations-based theory can produce the same predictions of alternative theories:

1. Status-quo: if expectations are stable, recent expectations will reflect the status quo
2. Lagged consumption: recent consumption will shape expectations
3. Goals: goals can shape (or be shaped by) expectations

16 The value function

These three phenomena: loss aversion, the reflection effect and reference dependence are reflected in the Prospect Theory value function in the following way:

1. The value function of Prospect Theory is defined on changes in wealth or welfare, rather than on final wealth levels. In other words, gains and losses are defined relative to a reference point. The reference point can be current wealth, a social or psychological status quo, or an expectation about the outcome.

Utility from an outcome depends on distance to the relevant reference level. A multi-millionaire may not accept a 50:50 bet to win $550, lose $500 because they are not comparing to their total wealth but relative to the reference point of the status quo.

2. Perceptions of both gains and losses are characterised by diminishing marginal sensitivity in either direction. Successive incremental changes have a smaller marginal impact.

This is similar to decreasing marginal utility to wealth of expected utility theory. The two differ in the baseline. In expected utility theory, the starting value is zero wealth, with increases from there decreasing in marginal utility. In prospect theory, the starting value is the reference point, with both increases and decreases having smaller marginal effect as they increase in magnitude.

3. Losses loom larger than gains. People feel more strongly about a loss than they do an equivalent gain. They are often willing to reject gambles with a materially positive expected value.

These phenomena result in the following famous figure (from Kahneman and Tversky (1979)) with:

1. The Value Function defined upon the variation from a reference point

2. A kink at the origin: losses count more than gains.

3. Diminishing sensitivity to further changes from the reference point in both directions

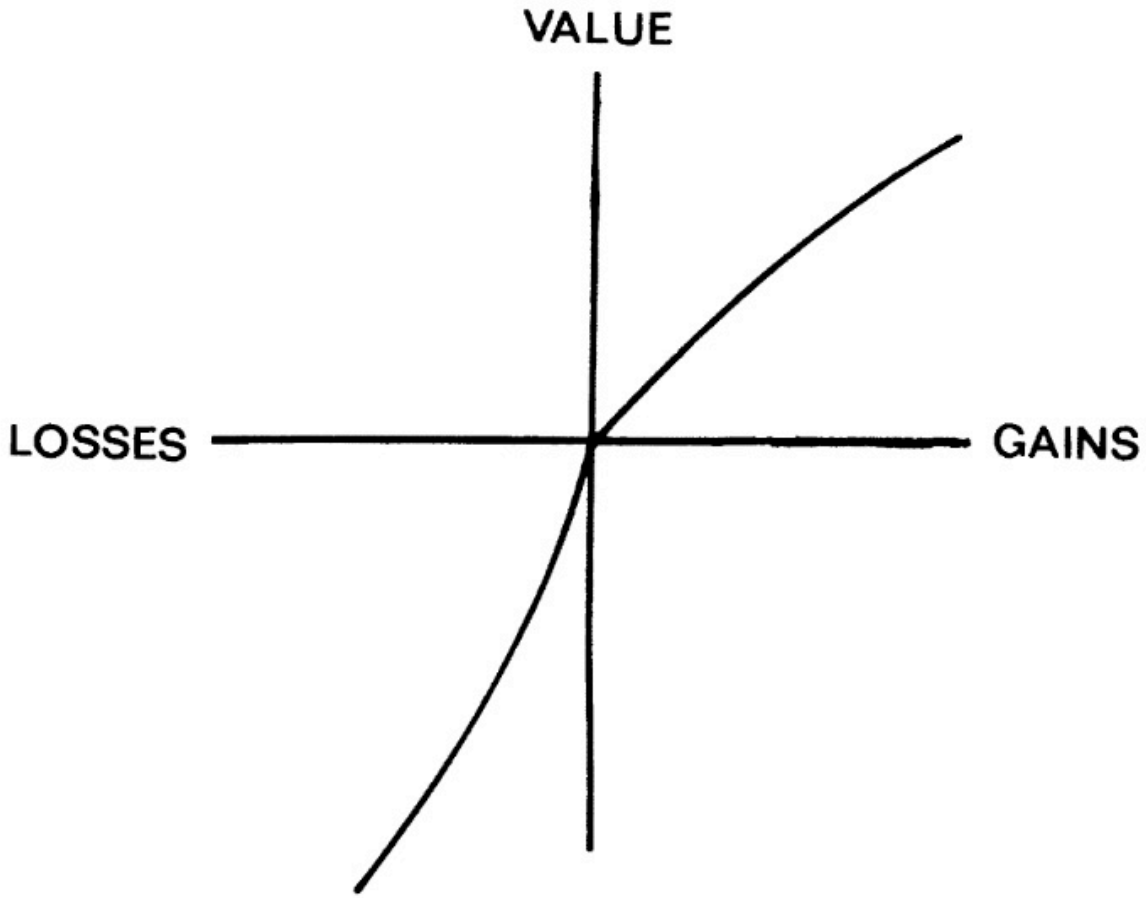


FIGURE 3.—A hypothetical value function.

17 Prospect theory implementation

Under Prospect theory, people evaluate the utility of a prospect by two phases: editing and evaluation.

17.1 Editing

Editing involves simplification of prospects for subsequent evaluation.

Kahneman and Tversky (1979) describes the editing phase as having four main operations:

- Coding: People perceive outcomes as gains or losses relative to some neutral reference position. The reference point is may be the status quo or some other expectation of agent.
- Combination: Prospects are simplified by combining probabilities for identical outcomes: (200, 0.25; 200, 0.25) will become (200, 0.50).
- Segregation: Riskless components are segregated out from risky components. For example (300, 0.80; 200, 0.20) corresponds to a a sure gain of 200 and the risky gamble (100, 0.80).
- Cancellation: Components that are shared by two prospects are ignored.

17.2 Evaluation

In the evaluation phase the prospects are evaluated and the option with the highest value chosen.

The value of a prospect is made up of:

1. a decision weight applied to each probability $\pi(p_i)$
2. the subjective value of each outcome $v(x_i)$

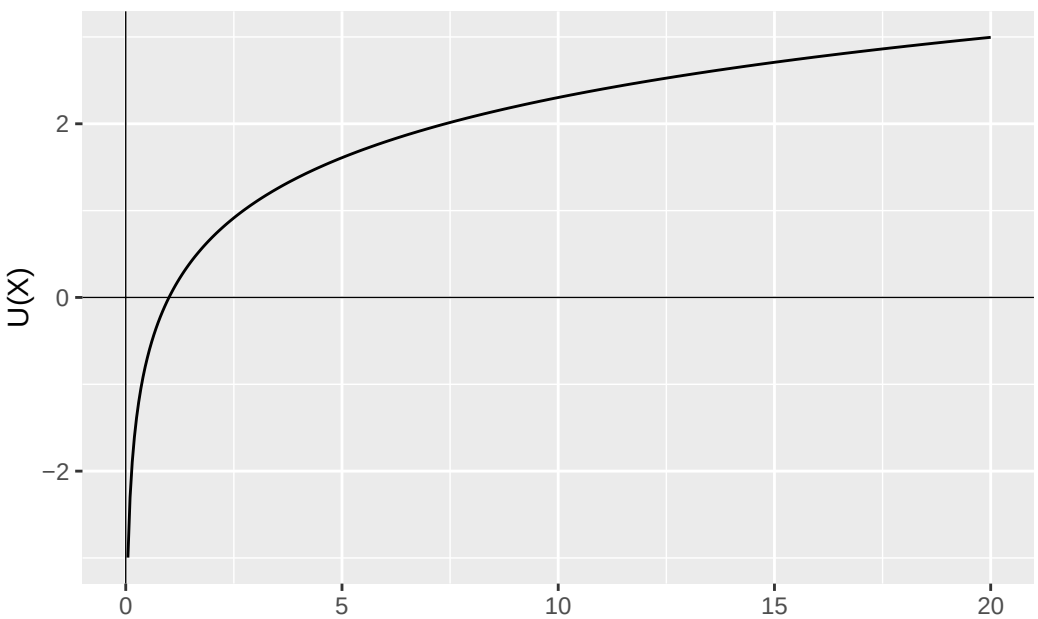
These are applied through the following formula (often called the value function):

$$
\begin{aligned}
V(X) &= \sum_{i=1}^{n} \pi(p_i) v(x_i) \\
&= \pi(p_1)v(x_1) + \pi(p_2)v(x_2) + ... + \pi(p_n)v(x_n)
\end{aligned}
$$

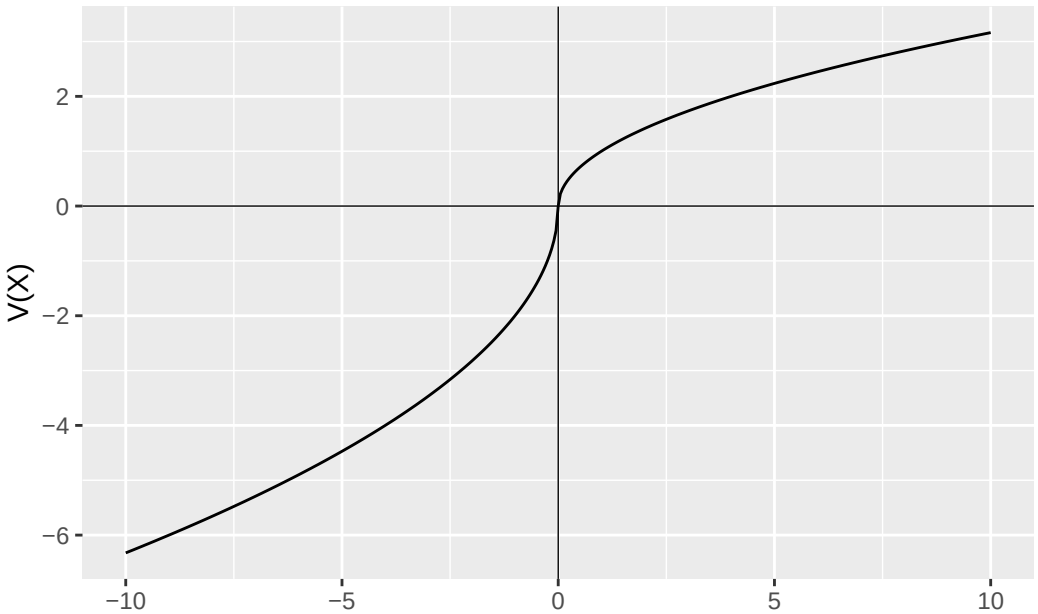
Where $V(X)$ is the expected value of the outcomes from gamble X.

The following diagrams illustrate the different shapes of the expected utility function and the Prospect Theory value function.

The expected utility function has diminishing marginal utility as utility increases. Utility is measured from a base reference point.



For the prospect theory function, you can see the kink at zero, with losses weighted more heavily than gains, with gains and losses determined relative to a reference point. There is diminishing sensitivity to further changes in both directions



17.3 Fourfold pattern of risk attitudes

Prospect theory results in a four-fold pattern of risk attitudes, as shown in this table. For moderate to high probability gambles, the reflection effect dominates and people are risk averse in the domain of gains and risk seeking in the domain of losses.

But for low probability gambles, the probability weighting shifts the decision calculus. The possibility of a gain is overweighted, making the gamble attractive and inducing risk seeking behaviour. A similar effect occurs for a low probability of loss, with the overweighted probability making the potential loss less attractive, inducing risk averse behaviour.

	Gains	Losses
Medium to high probability	Risk aversion	Risk seeking
Low probability (possibility effect)	Rick seeking	Risk aversion

18 Prospect Theory examples

18.1 Insurance

The classical economic explanation for the purchase of insurance is based on the risk aversion of consumers. Consumers are willing to buy insurance with a negative expected value as the consumer prefers the certainty of the premium payment to the risk of suffering an uninsured loss. (The negative expected value is due to the insurer's profit and administrative costs.)

Prospect theory provides an alternative explanation. The purchase of insurance involves a certain loss (the premium) or a gamble involving the possibility of either a large loss or the status quo. As prospect theory has people as risk seeking in the loss domain, we would not expect them to purchase insurance.

However, under prospect theory people also overweight small probabilities. This overweighting of small probabilities can make the purchase of insurance attractive even though it is in the loss domain. This combination of the loss domain but small probabilities is the bottom-right quadrant of the fourfold pattern to risk attitudes generated by prospect theory.

The following numerical example is an illustration.

An agent is considering insurance against bushfire for its \$1,000,000 house. The house has a 1 in 1000 chance of burning down. An insurer is willing to offer full coverage for \$1100. (Note: \$1000 is the actuarially fair price, the additional \$100 might represent profit or administrative costs.)

Would an expected value maximiser or risk neutral person purchase the insurance?

$$E[\text{purchase}] = -\text{premium} = -\$1,100$$

The expected value of purchasing insurance is the guaranteed loss of the premium.

$$\begin{aligned} E[\text{don't}] &= P_{\text{burn}} * -value_{\text{house}} \\ &= -0.001 * 1000000 \\ &= -\$1000 \end{aligned}$$

The expected value of purchasing insurance is \$100 less than the expected value of risking the house burning down. A risk neutral agent (who maximises expected value) would not purchase this insurance.

Would a risk averse agent purchase the insurance? Suppose they have a logarithmic utility function ($U(x) = ln(x)$) and they have \$10,000 in cash in addition to their house, giving them wealth (W) of \$1,010,000.

$$\begin{aligned} E[U(\text{purchase})] &= ln(W - premium) \\ &= ln(1,008,900) \\ &= 13.8244 \end{aligned}$$

$$\begin{aligned} E[U(\text{don't})] &= 0.999 * ln(W) + 0.001 * ln(W - value_{\text{house}}) \\ &= 0.999 * ln(1,010,000) + 0.001 * ln(10,000) \\ &= 13.8208 \end{aligned}$$

The expected utility of purchasing insurance is greater than the expected utility from not purchasing insurance. This agent will insure against the fire despite it being actuarially unfair.

Consider an agent who is risk seeking in the domain of losses but weights probability linearly. Their value function is:

$$v(x) = \begin{cases} x^{0.8} & \text{where} x \geq 0 \\ -2(-x)^{0.8} & \text{where} x < 0 \end{cases}$$

Where x is the realised outcome relative to the reference point.

Determination of the reference point can be arbitrary. What if you pay insurance every year? Could the reference point then be wealth minus the insurance payment (meaning the insurance payment is in the gain domain)?

Taking the reference point as current wealth, would this agent purchase the insurance?

$$\begin{aligned} V(purchase) &= v(-1,100) \\ &= -(1,100)^{0.8} \\ &= -271.1 \end{aligned}$$

$$
\begin{aligned}
V(don't) &= 0.999 * (0) + 0.001 * v(-1,000,000) \\
&= 0.999 * 0 - 0.001 * (1,000,000)^{0.8} \\
&= -63.1
\end{aligned}
$$

As $V(purchase) < V(don't)$, the agent does not purchase insurance. The diminishing feeling of loss leads to them weigh the certain loss of the premium relatively more heavily than the chance of losing the value of their house.

Including loss aversion in the value function does not change the decision as all possible outcomes are in the loss domain.

Would a person who is risk seeking in the domain of losses (i.e. the value function with reflection effect above) and applies the decision weights described below purchase the insurance?

They apply decision weights as per the following table:

Probability	0.001	0.01	0.1	0.25	0.5	0.75	0.9	0.99	0.999
Weight	0.01	0.05	0.15	0.3	0.5	0.7	0.85	0.95	0.99

$$
\begin{aligned}
V(purchase) &= v(-1,100) \\
&= -(1,100)^{0.8} \\
&= -271
\end{aligned}
$$

$$
\begin{aligned}
V(don't) &= \sum_{i=1}^{n} \pi(p_i) v(x_i) \\
&= \pi(0.999) * v(0) + \pi(0.001) * v(-1,000,000) \\
&= 0.99 * 0 - 0.01 * (1,000,000)^{0.8} \\
&= -631
\end{aligned}
$$

Although the diminishing feeling of loss leads to them weigh the certain loss of the premium relatively more heavily than the chance of losing the value of their house, the overweighting of the probability of fire leads them to purchase insurance. Again, if we had included loss aversion it would not have changed the decision as all possible outcomes are in the loss domain.

18.2 A gamble

Suppose an agent has the following reference-dependent utility function:

$$v(x) = \begin{cases} x^{1/2} & \text{where} x \geq 0 \\ -2(-x)^{1/2} & \text{where} x < 0 \end{cases}$$

Where x is the realised outcome relative to the reference point.

Assume that the agent's reference point is the status quo and the agent is offered the gamble A:

$$(\$110, 0.5; -\$100, 0.5)$$

Question 1

Will they want to play this gamble?

The utility from the gamble is:

$$\begin{aligned} V(A) &= 0.5 * v(110) - 0.5 * v(-100) \\ &= 0.5 * (110)^{0.5} - 0.5 * 2 * (100)^{0.5} \\ &= -4.76 \end{aligned}$$

They will not want to play this gamble as it has a negative value for the agent. They could receive value of 0 by simply not playing.

The reason for this negative value is that the agent is loss averse. The loss of \$100 is given twice the weight of an equivalent gain.

Question 2

Suppose the agent loses their wallet containing \$100. They feel bad about it and perceive it as a loss. Their reference point is unchanged at the original status quo, but the amount of money they have any after any outcome is \$100 less than otherwise. Would they be willing to take gamble A now?

After losing \$100 but not changing their reference point, they have two possible outcomes relative to their reference point: a gain of \$10 (winning \$110 minus the lost money in the wallet) and a loss of \$200 (losing \$100 and also losing their wallet).

$$
\begin{aligned}
V(A) &= 0.5 * v(-100 + 110) - 0.5 * v(-100 - 100) \\
&= 0.5 * (10)^{0.5} - 0.5 * 2 * (200)^{0.5} \\
&= -12.56
\end{aligned}
$$

The utility of not playing the gamble involves remaining with a loss of \$100:

$$
\begin{aligned}
V(\neg A) &= v(-100) \\
&= -2 * (100)^{0.5} \\
&= -20
\end{aligned}
$$

They will now want to play the gamble as it has a greater value than staying with their current loss. The reason the gamble becomes attractive is because it gives an opportunity to recover the loss. The agent is risk seeking over the loss domain. (They would even accept a 50:50 gamble to win \$100, lose \$100 with an expected value of zero.)

Question 3

The agent has now adapted to their loss of \$100. The new reference point is the new wealth level incorporating the loss wallet. Would they take gamble A now?

We are now back to an identical situation as in Question 1. They will not want to partake in the gamble.

Question 4

The agent wins \$10,000 at the casino. They feel good about their win, so their reference point remains at their wealth excluding the win. Would they take gamble A now?

With the additional \$10,000, the value from the gamble is now:

$$
\begin{aligned}
V(A) &= 0.5 * v(10000 + 110) + 0.5 * v(10000 - 100) \\
&= 0.5 * (10110)^{0.5} + 0.5 * (9900)^{0.5} \\
&= 100.02
\end{aligned}
$$

The value of not playing the gamble is:

$$
\begin{aligned}
V(\neg A) &= v(10000) \\
&= (10000)^{0.5} \\
&= 100
\end{aligned}
$$

The gamble is now attractive. They are less risk averse at a higher wealth, and the gamble is entirely in the gain domain, meaning that loss aversion does not affect the decision.

19 Prospect theory applications

19.1 Taxi drivers on rainy days

Camerer et al. (1997) studied the labour supply of New York City taxi drivers.

The taxi drivers in the study rent a cab for a 12-hour period for a fixed fee, plus petrol. Within the 12-hours, a driver can choose how long they keep the taxi out.

For many reasons, such as weather, subway breakdowns, day of the week and conferences, a taxi driver's effective wage can vary. When they are busier, they have a higher effective wage (that is, more fares).

Camerer et al. found that:

- The wage elasticity of drivers - that is, how their hours of driving respond to changes in their effective wages - was negative in two of the three samples. That is, drivers drove less when their effective wages were higher.
- For inexperienced drivers, elasticities were negative in all three samples and significantly greater in magnitude than that for experienced drivers.

Camerer et al. argue that this result is because taxi drivers have a one-day time horizon for decision making, whereby they have an earnings target beyond which they derive little additional utility. This leads them to work until they reach their target, which they reach more quickly on days with a higher wage.

- People have a tendency to engage in "narrow bracketing" when they make decisions, isolating them as single decisions (how much should I work today?) rather than thinking about them as a stream (how much should I work each day this week?)
- Aversion to falling below the reference point is consistent with loss aversion, with a result below the reference point causing stronger feelings than a result a similar amount above the reference point.

There have been numerous follow-up studies of taxi-drivers:

- Farber (2005) studied New York cabdrivers and found that the decision to stop work was primarily a function of how many hours had been worked up to that point in the day. He identified the difference between his and Camerer et al.'s result as being due to different empirical methods and measurement problems with the Camerer et al. data.

- Farber (2008) found that a model of labour supply with reference-dependent targets has a better fit than a standard neoclassical model, but that there was substantial variation day-to-day in any given driver's reference income level and that most shift ends before that reference income is reached.
- Farber (2015) used a much larger dataset on New York taxi driver behaviour and found that, as standard economic theory would predict, taxi drivers drive more when they can earn more. Farber also found that drivers did not earn more when it was raining.
- Martin (2017) examined taxi driver labour supply incorporating the S-shaped reference dependence of Prospect Theory (the reflection effect with risk seeking behaviour in the loss domain and risk aversion in the gain domain. Martin found evidence that taxi driver behaviour was consistent with this fully form of Prospect Theory. He differentiated from the other papers on the basis that they considered a narrower version of reference dependence focusing on loss aversion only.

19.2 The disposition effect

The disposition effect is the tendency for investors to sell stocks that are in the gain domain relative to the purchase price and to hold stocks that are in the loss domain (Shefrin and Statman (1985)).

While tax implications or portfolio rebalancing are both potential explanations for asymmetric behaviour relating to the sale of stocks, these factors have been shown to be insufficient to explain the observed behaviour.

Most behavioural explanations have turned to prospect theory.

For example, Shefrin and Statman (1985) argued that the disposition effect is driven by the reflection effect, whereby investors are risk seeking in the loss domain and risk averse in the gain domain. To demonstrate how it works, they present the following scenario:

> [C]onsider an investor who purchased a stock one month ago for $50 and who finds that the stock is now selling at $40. The investor must now decide whether to realize the loss or hold the stock for one more period. To simplify the discussion, assume that there are no taxes or transaction costs. In addition, suppose that one of two equiprobable outcomes will emerge during the coming period: either the stock will increase in price by $10 or decrease in price by $10. According to prospect theory, our investor frames his choice as a choice between the following two lotteries:
>
> A. Sell the stock now, thereby realizing what had been a $10 "paper loss".
>
> B. Hold the stock for one more period, given 50-50 odds between losing an additional $10 or "breaking even."

For an investor who is risk seeking in the loss domain, option B would be attractive.

If we craft an alternative scenario where the stock is now selling at $60, selling would realise a $10 gain, while holding the stock would be a risky prospect with the same expected value. An investor who is risk averse in the gain domain will sell.

19.3 The housing market

Genesove and Mayer (2001) examined hosing data from Boston. They found that owners subject to nominal losses:

- Set higher asking prices of 25-35% of the difference between the expected selling price and their original purchase price
- Attain higher prices of 3-18% of that difference

This suggests sellers are averse to realising nominal losses.

Part IV

Intertemporal choice

Intertemporal choice refers to decisions involving costs and benefits occurring at different times.

Almost every decision is an intertemporal choice. Intertemporal choices can be important because:

- feedback may not be immediate (when will you realise your retirement savings are inadequate?)
- they may be irreversible (what can you do if you reach retirement age with little savings?)
- stakes can be large (health, wealth, career).

Notation

We have seen preferences over lotteries of the form:

$$L = (x_1, p_1; x_2, p_2; ...; x_n, p_n)$$

We are now going to see preferences over stream of payoffs of the form:

$$S = (t_1, x_1; t_2, x_2; ...; t_n, x_n)$$

An example of a stream of income is a salary: \$50,000 this year, \$50,000 next year, and so on.

Let's consider a two simple streams of payoffs. The first stream is \$100 now and the second stream is \$107 in one year. We can write these two streams as:

$$S_1 = (0, \$100; 1, 0)$$

$$S_2 = (0, 0; 1, \$107)$$

Period $t_1 = 0$ is now. Period $t_2 = 1$ is one year from today.

Time preference

Would you prefer S_1 or S_2?

People tend to discount future costs and benefits. They prefer to receive benefits earlier, rather than later, and prefer to incur costs later rather than earlier.

20 Exponential discounting

Exponential discounting occurs where future costs and benefits are discounted at a consistent rate through time. The following equation is an example of exponential discounting.

$$
\begin{aligned}
U_0 &= u(x_0) + \delta u(x_1) + \delta^2 u(x_2) + \delta^3 u(x_3) + ... + \delta^T u(x_T) \\
&= \sum_{t=0}^{t=T} \delta^t u(x_t)
\end{aligned}
$$

$$0 \leq \delta \leq 1$$

where U_0 is utility at time $t = 0$, and x_t is the payoff in period t. δ is a discount factor reflecting how much the decision maker weights the future relative to today.

The degree of discounting in this equation evolves over time as 1, δ, δ^2, δ^3, δ^4 and so on. This results in a smooth decline in present value over time.

δ is the discount factor. The higher the discount factor, the less the agent discounts future costs and benefits. You will often see discussion of the "discount rate", r. In discrete time, the relationship between δ and r is as follows:

$$\delta = \frac{1}{1+r}$$

A larger discount factor implies less discounting. A larger discount rate implies more discounting.

20.1 Exponential discounted utility model assumptions

Under the standard model of exponential discounting, at =0 the agent seeks to maximize the utility of the future path of consumption. It is underpinned by the following assumptions:

20.1.1 Time-consistency

Once the agent starts moving along the consumption path, they are time-consistent with their own initial plan. For example, consider an agent facing the following two choices:

Would you like \$100 today or \$110 next week?

Would you like \$100 next week or \$110 in two weeks?

An exponential discounter will choose \$100 in both choices or \$110 in both choices. The reason is that after one week the second choice effectively becomes the same as the first choice. Time consistency implies that they will continue to want to make the same choice regardless of when they are making it.

20.1.2 Consumption independence

Consumption independence means that utility in period $t+k$ is independent of consumption in any other period. An outcome's utility is unaffected by outcomes in prior or future periods.

Imagine a world with the consumption good x = a behavioural economics subject.

An exponential discounted utility maximiser wants to consume Lecture 1 at $t+3$ and Lecture 2 at $t+4$. Under consumption independence, if the agent does not attend Lecture 1, they still expect to benefit from Lecture 2 at $t+4$ consistent with the plan they decided at t.

This assumption allows us to write $x = x_1 + x_2 + x_3 + ... + x_n$. That is, good x can be split and allocated across periods.

20.1.3 Stationary preferences

$U_t = U_{t=K}$. The utility function is stationary across periods. The functional form of U_t is the same as the functional form of U_{t+k}.

20.1.4 Utility independence

Under utility independence, all that matters is maximizing the sum of discounted utilities. Decision makers have no preference for the distribution of utilities.

21 Exponential discounting examples

21.1 Example 1

Suppose we have an exponential discounter with $\delta = 0.95$ and utility each period of $u(x_n) = x_n$.

Choice 1: Would this agent prefer \$100 today ($t = 0$) or \$110 next week ($t = 1$)?

$$\begin{aligned} U_0(0, \$100) &= u(\$100) \\ &= 100 \end{aligned}$$

$$\begin{aligned} U_0(1, \$110) &= \delta u(\$110) \\ &= 0.95 * 110 \\ &= 104.50 \end{aligned}$$

The exponential discounter will prefer to receive \$110 next week.

Choice 2: Would this agent prefer \$100 next week ($t = 1$) or \$110 in two weeks ($t = 2$)?

$$\begin{aligned} U_1(1, \$100) &= \delta u(\$110) \\ &= 0.95 * 100 \\ &= 95 \end{aligned}$$

$$\begin{aligned} U_1(2, \$110) &= \delta^2 u(\$110) \\ &= 0.95^2 * 110 \\ &= 99.275 \end{aligned}$$

The exponential discounter will prefer to receive \$110 in two weeks. The set of decisions across Choice 1 and Choice 2 are time consistent. If the agent selected \$110 in two weeks for Choice 2 and was given a chance to change their choice after one week (which is effectively Choice 1), they would not change.

21.2 Example 2

Suppose we have an exponential discounter with $\delta = 0.95$ per week and utility each period of $u(x_n) = x_n$

They are offered \$100 today. What sum would they need to be offered in one year (52 weeks) to prefer that later payment to the \$100 today?

$$\begin{aligned} U_0(0, \$100) &= u(\$100) \\ &= 100 \end{aligned}$$

$$\begin{aligned} U_0(52, \$x) &= \delta^{52} u(\$x) \\ &= 0.95^{52} * x \end{aligned}$$

They will prefer $\$x$ in 52 weeks if $U(\$x \text{ at } t = 52)$ is greater than 100.

$$\begin{aligned} 0.95^{52} * x &> 100 \\ x &> \frac{100}{0.95^{52}} \\ x &> \$1440.03 \end{aligned}$$

The agent would be willing to wait a year for payment if they were paid more than \$1440.03.

22 Exponential discounting anomalies

22.1 Estimation of δ

Frederick et al. (2002) plotted estimates of from a set of published papers. Figure 1a shows that the estimated discount factor increases with time horizon. If studies with horizons of one year or less are excluded, there is no relationship between the discount factor and time horizon (Figure 1b).

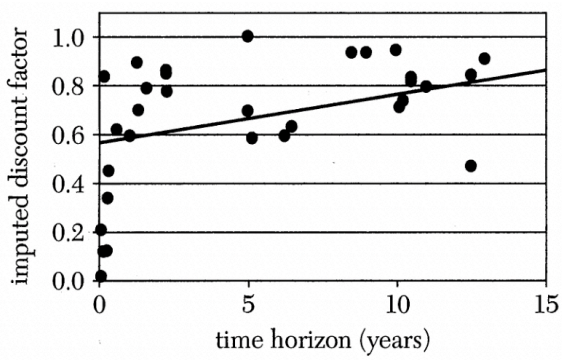


Figure 1a. Discount Factor as a Function of Time Horizon (all studies)

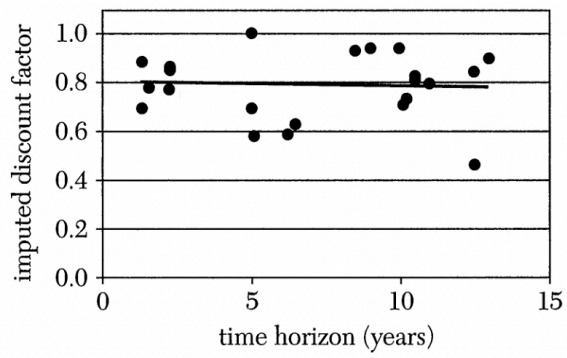


Figure 1b. Discount Factor as a Function of Time Horizon (studies with avg. horizons > 1 year)

Thaler (1981) estimated the discount rate of experimental subjects by presenting them with a choice between a prize today or a larger later. In each case, they were asked, given the size of the prize that could be received today, how large would the future prize would need to be such that waiting would be as attractive as receiving the money now.

For example, some subjects were asked how large a future prize would need to be such that they would be happy to wait three months, one year or three years rather than receiving \$15 today. The median answers were \$30, \$60 and \$100 respectively. If you use these estimates to calculate an implied discount rate, the result is 277%, 139% and 63%. It can be seen that this discount rate is decreasing with the length of delay (which matches the pattern of increasing discount factor with length of delay).

22.2 Preference reversal

Green et al. (1994) offered experimental subjects choices between a smaller reward and delayed larger reward while varying the delay. For example, they offered:

- $20 now or $50 in three months
- $20 in one week or $50 in three months and one week.

Across the choices offered to the experimental subjects, there was a consistent effect whereby incrementing the delay for both rewards equally would result in a switch from the small sooner reward to the latter larger reward.

In a study by Kirby and Herrnstein (1995), 34 of 36 experimental subjects reversed preference from a. Larger later reward to a smaller earlier reward as the delays to both decreased.

22.2.1 Healthy choices

Read and Leeuwen (1998) asked 200 study participants to make a choice between healthy or unhealthy snacks (e.g., banana or apple versus Mars bar or Snickers bar) that they would receive in one week. 48% of men and 51% of women chose the healthy choice.

At the scheduled time one week later they asked the participants to choose again. Although no reference was made to their previous choice, this effectively gave them the opportunity to change their mind. This time, only 25% of men and 11% of women chose the healthy snack. While many changed from the healthy to unhealthy snack, almost no changed from the unhealthy to healthy snack.

This result is evidence of time inconsistency. The participants made different decisions depending upon when they made the decision.

23 Present bias

Present bias occurs when we place additional weight on costs and benefits at the present time.

23.1 The $\beta\delta$ model

One simple model of present bias is the quasi-hyperbolic discounting model (it is a discrete time version of hyperbolic discounting).

$$\begin{aligned} U_0 &= u(x_0) + \beta\delta u(x_1) + \beta\delta^2 u * (x_2) + ... + \beta\delta^T u(x_T) \\ &= u(x_0) + \beta \sum_{t=1}^{t=T} \delta^t u(x_t) \end{aligned}$$

$0 \leq \delta \leq 1$

$0 \leq \beta \leq 1$

β is the short-term discount factor: it scales down all future dates utility.

δ is the usual discount factor.

The degree of discounting in this equation evolves over time as 1, $\beta\delta$, $\beta\delta^2$, $\beta\delta^3$, $\beta\delta^4$ and so on. This progression results in a larger discount for the first period of delay ($\beta\delta$) than the degree of discount for each subsequent period of delay (δ). There is a relative weighting toward the present. Present bias of this nature can result in time inconsistency, with decisions at one point reversed at another if given the opportunity.

23.2 Assumptions

Under the $\beta\delta$ model we are loosening the assumption of time consistency. An agent may change their initial plan through time. However, we are maintaining many of the other assumptions that we used when we examined exponential discounting. We are maintaining the assumptions of:

- Consumption independence: Utility in period $t+k$ is independent of consumption in any other period. An outcome's utility is unaffected by outcomes in prior or future periods.
- Stationary preferences: $U_t = U_{t+k}$. That is, the utility function is stationary across periods, that is: functional form of U_t = functional form of U_{t+1}, U_{t+2} etc.
- Utility Independence: All that matters is maximizing the sum of discounted utilities. Decision makers are assumed to have no preference for the distribution of utilities.

24 Present bias examples

24.1 Example 1

Choice 1: Would you like \$100 today or \$110 next week?

Choice 2: Would you like \$100 next week or \$110 in two weeks?

24.1.1 The exponential discounter

Suppose we have an exponential discounter with $\delta = 0.95$ and utility each period of $U(x_n) = x_n$.

Choice 1: Would this agent prefer \$100 today ($t = 0$) or \$110 next week ($t = 1$)?

We have already worked through this problem and determined that an exponential discounter will prefer to receive \$110 next week.

Choice 2: Would this agent prefer \$100 next week ($t = 1$) or \$110 in two weeks ($t = 2$)?

Similarly, we previously determined that the exponential discounter will prefer to receive \$110 in two weeks. The set of decisions across Choice 1 and Choice 2 are time consistent.

24.1.2 The present-biased agent

Suppose we have a present biased agent with $\delta = 0.95$, $\beta = 0.95$ and utility each period of $U(x_n) = x_n$.

Choice 1: Would this agent prefer \$100 today ($t = 0$) or \$110 next week ($t = 1$)?

$$\begin{aligned} U_0(0, \$100) &= u(\$100) \\ &- 100 \end{aligned}$$

$$
\begin{aligned}
U_0(1, \$110) &= \beta\delta u(\$110) \\
&= 0.95 * 0.95 * 110 \\
&= 99.275
\end{aligned}
$$

As $U_0(0, \$100) > U_0(1, \$110)$, the present-biased agent will prefer to receive \$100 this week.

Choice 2: Would this agent prefer \$100 next week ($t = 1$) or \$110 in two weeks ($t = 2$)?

$$
\begin{aligned}
U_0(1, \$100) &= \beta\delta u(\$100) \\
&= 0.95 * 0.95 * 100 \\
&= 90.25
\end{aligned}
$$

$$
\begin{aligned}
U_0(2, \$110) &= \beta\delta^2 u(\$110) \\
&= 0.95 * 0.95^2 * 110 \\
&= 94.31
\end{aligned}
$$

As $U_0(1, \$100) < U_0(2, \$110)$, the present-biased agent will prefer to receive \$110 in two weeks.

Putting those two choices together:

Choice 1: The present-biased agent will prefer \$100 now to \$110 in one week. Their preference for benefits now (β) leads them to prefer the immediate payoff.

Choice 2: The present-biased agent will prefer \$110 in two weeks to \$100 in one week. They are willing to wait longer for a larger reward, with both outcomes in the future and subject to the short-term discount (β).

Consider what would happen if they selected the \$110 in two weeks in Choice 2, but after one week were asked if they would like to reconsider their choice. They are effectively being offered Choice 1. This would then lead them to change their mind and take the immediate \$100.

This combination of decisions is time inconsistent. The agent's actions are not consistent with their own initial plan.

25 Sophisticated present bias

25.1 Naïve and sophisticated present-biased agents

A naive present-biased agent discounts all future periods but incorrectly believes that they do not have this problem in the future.

A sophisticated present-biased agent discounts all future periods and correctly believes that they will have this problem also in the future. They understand that if faced with the temptation to take benefits today or delay costs, they will do so.

A sophisticated and naive person can make different decisions despite having the same level of impatience, δ, and the same level of present bias, β.

The naive agent makes its plans by forward reasoning, starting from today.

1. Solve for the optimal action given the possible futures, believing they will stick to the plan once they move to next period.
2. Move to next time period $(t + 1)$
3. Recompute their optimal plan, believing they will stick to the plan once they move to next period.
4. Iterate

The sophisticated agent makes its plans by backward reasoning, starting from the final time period.

1. Solve for the optimal action
2. Go back one time period and solve for preferred action, accounting for the decision in (1)
3. Iterate

25.2 Examples

25.2.1 Example 1

Suppose we have a naive and a sophisticated present-biased agent, each with $\delta = 0.95$, $\beta = 0.95$ and utility each period of $U(x_n) = x_n$.

We offer them the following choice.

> Would you like \$100 next week or \$110 in two weeks?

We also tell them that we will give them the opportunity to reconsider their decision next week.

The present-based agent that we examined in the earlier example was naïve. They did not realise that they would change their mind if offered that opportunity after a week. They would be surprised by that change.

When making their initial decision, the comparison they are making from the perspective of today is simply between the discounted utilities from the perspective of today:

$$U_0(1, \$100) = 90.25$$

$$U_0(2, \$110) = 94.32$$

But this is an incorrect view as to their preferences next week. Next week they will calculate utility as:

$$U_1(1, \$100) = 100$$

$$U_1(2, \$100) = 99.275$$

Today they hold an incorrect belief as to what their preference will be next week .

The sophisticated agent will first consider how they will look at the choice next week:

$$U_1(1, \$1001) = 100$$

$$U_1(2, \$110) = 99.275$$

The sophisticated agent sees that next week they will take the \$100.

As a result, today the sophisticated agent will choose \$100 in one week. They see their future decision and know that \$110 in two weeks is not available to them. They effectively accept their future present bias now.

In this example, being naive or sophisticated does not change their final decision. It only changes their beliefs about their final decision before they reach it.

25.2.2 Example 2

Suppose we have a naive and a sophisticated present-biased agent, each with $\delta = 1$, $\beta = 0.5$ and utility each period of $U(x_n) = x_n$. They are present-biased, but beyond that present-bias demonstrate no impatience.

We offer them the following choice.

- \$6 today ($t = 0$)
- \$10 in one week ($t = 1$)
- \$16 in two weeks ($t = 2$)

They are also told that next week they will be offered an opportunity to change their mind.

[Note: I am using \$ for many of these examples, but these could equally be many other things: e.g. three movies that each provide 6, 10 or 16 utility and you can only afford to attend one.]

First, we consider the naive agent. They calculate utility from the perspective of today.

$$\begin{aligned} U_0(0, \$6) &= u(\$6) \\ &= 6 \end{aligned}$$

$$\begin{aligned} U_0(1, \$10) &= \beta\delta u(\$10) \\ &= 0.5 * 1 * 10 \\ &= 5 \end{aligned}$$

$$\begin{aligned} U_0(2, \$16) &= \beta\delta^2 u(\$16) \\ &= 0.5 * 1^2 * 16 \\ &= 8 \end{aligned}$$

The naive agent will choose the \$16 in two weeks.

But what then happens when the naive agent is given the chance to change their mind after one week.

$$\begin{aligned} U_1(\$10 \text{ at } t = 1) &= u(\$10) \\ &= 10 \end{aligned}$$

$$\begin{aligned} U_1(\$16 \text{ at } t = 2) &= \beta\delta u(\$16) \\ &= 0.5 * 1 * 16 \\ &= 8 \end{aligned}$$

The naive agent will change their mind and take the \$10.

What of our sophisticated agent? They will make their decision today based on their correct beliefs about their future preferences. To do this, they solve by backward induction. First, what will their decision be next week?

$$\begin{aligned} U_1(1, \$10) &= u(\$10) \\ &= 10 \end{aligned}$$

$$\begin{aligned} U_1(2, \$16) &= \beta\delta u(\$16) \\ &= 0.5 * 1 * 16 \\ &= 8 \end{aligned}$$

The sophisticated agent can see that they will choose to receive \$10 immediately when asked next week.

Knowing this is the case, the sophisticated agent now decides whether they prefer the \$6 today or the \$10 next week.

$$\begin{aligned} U_0(0, \$6) &= u(\$6) \\ &= 6 \end{aligned}$$

$$
\begin{aligned}
U_0(1, \$10) &= \beta\delta u(\$10) \\
&= 0.5 * 1 * 10 \\
&= 5
\end{aligned}
$$

They prefer the \$6 today.

The result is that the sophisticated agent will choose the \$6 today. From today's perspective, they would prefer the \$16 in two weeks, but as they know they will cave in to their present bias next week and take the \$10, they make today's decision on that basis.

25.2.3 Example 3

A naive present-biased agent has failed to return their library books and is fined at at $t = 0$. They can select one of the following payment schemes:

$(0, -\$10)$, $(1, -\$15)$ or $(2, -\$25)$

That is, they can pay \$10 today, \$15 at $t = 1$ or \$25 at $t = 2$.

They are free to change the scheme over time as they see fit.

The agent's utility is linear in money $U(x_n) = x_n$, with discount factors $\delta = 1$ and $\beta = 0.5$.

We start calculating utility from today:

$$
\begin{aligned}
U_0(0, -\$10) &= u(-\$10) \\
&= -10
\end{aligned}
$$

$$
\begin{aligned}
U_0(1, -\$15) &= \beta\delta u(-\$15) \\
&= 0.5 * 1 * (-15) \\
&= -7.5
\end{aligned}
$$

$$
\begin{aligned}
U_0(2, -\$25) &= \beta\delta^2 u(-\$25) \\
&= 0.5 * 1^2 * (-25) \\
&= -12.5
\end{aligned}
$$

The naive agent will choose to pay \$15 at $t = 1$.

Now we move to $t = 1$, when the naive agent reconsiders their decision:

$$\begin{aligned} U_1(1, -\$15) &= u(-\$15) \\ &= -15 \end{aligned}$$

$$\begin{aligned} U_1(2, -\$25) &= \beta\delta u(-\$25) \\ &= 0.5 * 1 * (-25) \\ &= -12.5 \end{aligned}$$

The naive agent will choose to change their decision and further delay their payment. They will now choose to pay \$25 at $t = 2$.

When they reach $t = 2$, they have no choice but to pay the \$25.

A sophisticated present-biased agent has failed to return their library books and is fined at t =0. They can select one of the following payment schemes:

$(0, -\$10)$, $(1, -\$15)$ or $(2, -\$25)$

That is, they can pay \$10 today, \$15 at $t = 1$ or \$25 at $t = 2$.

They are free to change the scheme over time as they see fit.

The agent's utility is linear in money $U(x_n) = x_n$, with discount factors $\delta = 1$ and $\beta = 0.5$.

For the sophisticated agent, we start calculating utility from the final period.

At $t = 2$, they have no choice but to pay the \$25.

What of $t = 1$?

$$\begin{aligned} U_1(1, -\$15) &= u(-\$15) \\ &= -15 \end{aligned}$$

$$\begin{aligned} U_1(2, -\$25) &= \beta\delta u(-\$25) \\ &= 0.5 * 1 * (-25) \\ &= -12.5 \end{aligned}$$

The sophisticated agent can see that if they make a choice at $t = 1$, they will choose to pay \$25 at t = 2.

Now we iterate at t = 0. The sophisticated agent only compares \$10 at $t = 0$ with \$25 at $t = 2$ because they know that at $t = 1$ they will select \$25 at $t = 2$. They know that if they delay once they will delay again and end up paying the largest fine. Hence they limit their comparison to those outcomes that might occur:

$$\begin{aligned} U_0(0, -\$10) &= u(-\$10) \\ &= -10 \end{aligned}$$

$$\begin{aligned} U_0(2, -\$25) &= \beta\delta^2 u(-\$25) \\ &= 0.5 * 1^2 * (-25) \\ &= -12.5 \end{aligned}$$

As $U_0(0, -\$10) > U_0(2, -\$25)$, the sophisticated agent will choose to pay \$10 at $t = 0$.

In our examples we have seen two contrasting outcomes for the sophisticated agent.

In one example they took \$6 today (rather than \$10 in one week or \$16 in two) because they knew that they would take the \$10 in one week. They took the lowest possible outcome.

In the latest example, they paid the lowest possible fine as they saw they would further delay paying the fine in the future, leading it to increase even more.

Suboptimality of sophisticated behaviour arises from two tensions:

1. They understand that they will not be able to wait for the optimal time.
2. They are present-biased so prefer benefit today and costs delayed to the future.

In combination, these imply a sophisticated agent will generally take action earlier than the naive agent. They can "prepoperate", which is doing something now when it would be better to wait.

26 Commitment

A commitment device is a mechanism which locks yourself into a course of action that you might not otherwise choose but that produces the desired result.

Commitment devices may work through the following channels:

- Artificially depressing the value of the bad course of action
- Artificially increasing the value of the optimal course of action
- Forcing the agent to maintain the optimal course of action.

26.1 Commitment examples

26.1.1 Forcing the optimal course of action

Recall what the sophisticated agent did when faced with a choice of (0, \$6; 1, \$10; 2, \$16). First they realised that next week they would take the \$10 straight away.

$$\begin{aligned} U_1(1, \$10) &= u(\$10) \\ &= 10 \end{aligned}$$

$$\begin{aligned} U_1(2, \$16) &= \beta\delta u(\$16) \\ &= 0.5 * 1 * 16 \\ &= 8 \end{aligned}$$

Knowing this is the case, they then chose to take the \$6 today.

$$\begin{aligned} U_0(0, \$6) &= u(\$6) \\ &= 6 \end{aligned}$$

$$
\begin{aligned}
U_0(1, \$10) &= \beta\delta u(\$10) \\
&= 0.5 * 1 * 10 \\
&= 5
\end{aligned}
$$

But when they compare all three, they would prefer the \$16 in two weeks. It is only because they can foresee their future failing after one week that they don't wait.

$$
\begin{aligned}
U_0(0, \$6) &= u(\$6) \\
&= 6
\end{aligned}
$$

$$
\begin{aligned}
U_0(1, \$10) &= \beta\delta u(\$10) \\
&= 0.5 * 1 * 10 \\
&= 5
\end{aligned}
$$

$$
\begin{aligned}
U_0(2, \$16) &= \beta\delta^2 u(\$16) \\
&= 0.5 * 1^2 * 16 \\
&= 8
\end{aligned}
$$

But what if they could commit themselves today?

Suppose that not only could they select an option today, but they could also choose to bind themselves to that decision such that they could not change it in the future. The result of that: a sophisticated present-biased agent would choose the \$16 in two weeks, and bind their future self from changing their mind.

$$U_0(0, \$6) = 6$$

$$U_0(0, \$10) = 5$$

$$U_0(0, \$16) = 6$$

26.1.2 Forcing the optimal course of action: Odysseus

Another example of forcing the agent to maintain the optimal course of action is the story of Odysseus.

Odysseus was required to sail past the sirens at $t = 1$. At that time he can either jump off his ship to join the sirens (and die) or sail on past and live.

As Odysseus is a sophisticated present-biased agent, he considers what he is likely to do at that time. He realises that he will jump of the ship for the immediate benefit of joining the sirens at the loss of the longer-term benefit of living.

But from the perspective of Odysseus today, with both the benefits of the siren and living in the future, Odysseus would prefer to live. As a result, he decides to commit himself to that course of action by instructing his crew to tie him to the mast (plus leaving his ears unplugged so that he also gets some benefits from the siren song).

Figure 26.1: John William Waterhouse, Ulysses and the Sirens (1891)

26.1.3 Forcing the optimal course of action: lay-by

Lay-by involves paying a deposit and administrative fee toward the purchase of a product. You receive your purchase when you make payment in full some time later.

Let us see how that could help a quasi-hyperbolic discounter with discount factors $\delta = 1$ and $\beta = 1/2$.

Suppose you want a new jacket for work. You need to save for three months to purchase the jacket. But each month you save you forgo consumption that would boost your utility. Today you decide whether you will save for the jacket or not.

You receive the following payoffs for each course of action:

	t=0	t=1	t=2
Save	0	0	45
Spend	10	10	10

At $t = 0$ the highest discounted utility is for Save. You plan to save for the jacket.

Save:

$$\begin{aligned} U(x) &= 0 + \beta\delta * 0 + \beta\delta^2 * 45 \\ &= 0.5 * 45 \\ &= 22.5 \end{aligned}$$

Spend:

$$\begin{aligned} U(x) &= 10 + \beta\delta * 10 + \beta\delta^2 * 10 \\ &= 10 + 0.5 * 10 + 0.5 * 10 \\ &= 20 \end{aligned}$$

One month now passes. You have saved for a month. You could now spend both your savings from last month and this month, giving a short term boost, or keep saving for your jacket.

	t=1	t=2
Save	0	45
Spend	20	10

At $t = 1$ spending now has the highest discounted utility. After saving for the first period, you spend.

Save:

$$
\begin{aligned}
U(x) &= 0 + \beta\delta * 45 \\
&= 0.5 * 45 \\
&= 22.5
\end{aligned}
$$

Spend:

$$
\begin{aligned}
U(x) &= 20 + \beta\delta * 10 \\
&= 20 + 0.5 * 10 \\
&= 25
\end{aligned}
$$

You spend despite initially wanting to save.

Now let's consider this problem from the point of view of a sophisticated present-biased agent. They see their full choice set as:

	t=0	t=1	t=2
Save	0	0	45
Start spending at t=1	0	20	10
Spend	10	10	10

You work backward through time. At $t = 2$, you will buy the jacket.

The utility of each option at $t = 1$ is as follows:

Save:

$$
\begin{aligned}
U(x) &= 0 + \beta\delta * 45 \\
&= 0.5 * 45 \\
&= 22.5
\end{aligned}
$$

Spend:

$$
\begin{aligned}
U(x) &= 20 + \beta\delta * 10 \\
&= 20 + 0.5 * 10 \\
&= 25
\end{aligned}
$$

If you had spent at $t = 0$, you have no choice but to spend at $t = 2$.

The sophisticated quasi-hyperbolic discounter now knows that saving for the jacket is not available to them as they will spend at $t = 1$ regardless of their initial actions. As a result, at $t = 0$ they choose between the two feasible options:

Start spending at t=1:

$$\begin{aligned} U(x) &= 0 + \beta\delta * 20 + \beta\delta^2 * 10 \\ &= 0.5 * 20 + 0.5 * 10 \\ &= 15 \end{aligned}$$

Spend:

$$\begin{aligned} U(x) &= 10 + \beta\delta * 10 + \beta\delta^2 * 10 \\ &= 10 + 0.5 * 10 + 0.5 * 10 \\ &= 20 \end{aligned}$$

They start to spend at $t = 0$. Contrast this with the naïve agent who chooses Save at $t = 0$. The sophisticated agent would also prefer Save at $t = 0$, but knows that Save is not available to them as they will spend at $t = 1$.

Now consider what the sophisticated agent may do in the presence of lay-by.

For payment of an administrative fee equivalent to 1 unit of utility and an initial deposit (your savings in $t = 0$), you reserve the jacket, preventing you from spending that money in period 2. The new set of options is:

	t=0	t=1	t=2
Save	0	0	45
Start spending at t=1	0	20	10
Spend	10	10	10
Lay-by	-1	0	45

The lay-by option is strictly worse than Save. But what happens if lay-by is available to the sophisticated present-biased agent?

Again, working backward, at $t = 2$, the agent buys the jacket.

At $t = 1$, by the same logic we looked at previously, they spend if they are able to do so. That eliminates Save from their choice set. At \$ $t = 0$, they now compare the feasible options:

Start spending at t=1:

$$\begin{aligned} U(x) &= 0 + \beta\delta * 20 + \beta\delta^2 * 10 \\ &= 0.5 * 20 + 0.5 * 10 \\ &= 15 \end{aligned}$$

Spend:

$$\begin{aligned} U(x) &= 10 + \beta\delta * 10 + \beta\delta^2 * 10 \\ &= 10 + 0.5 * 10 + 0.5 * 10 \\ &= 20 \end{aligned}$$

Lay-by:

$$\begin{aligned} U(x) &= -1 + \beta\delta * 0 + \beta\delta^2 * 45 \\ &= -1 + 0.5 * 0 + 0.5 * 45 \\ &= 21.5 \end{aligned}$$

Lay-by is the preferred option at $t = 0$. As it binds the agent in the future, they are able to stick to this plan.

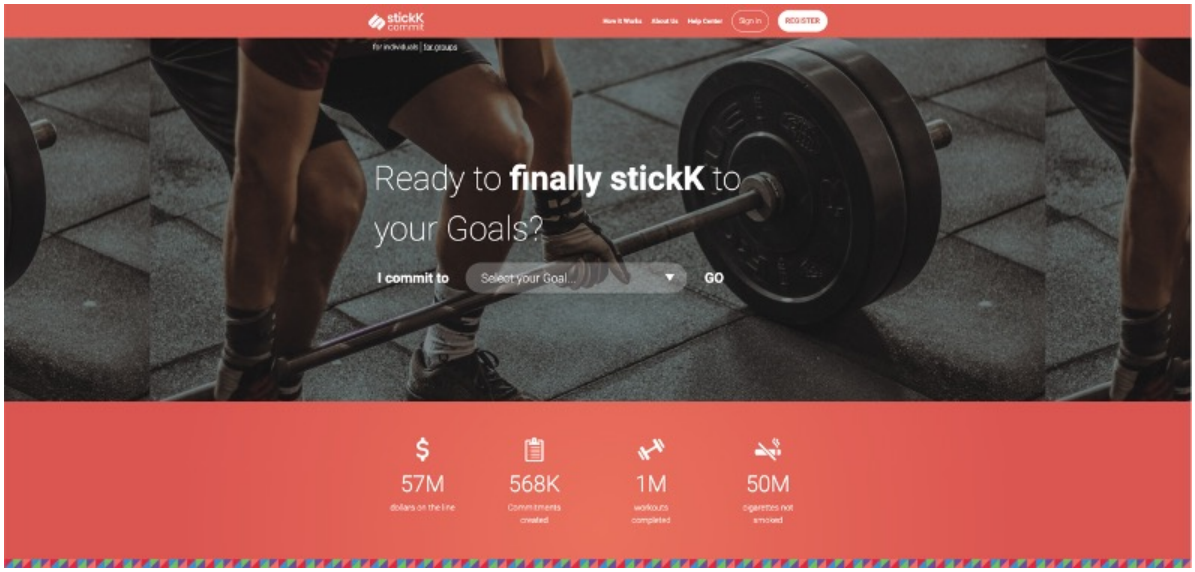
Note that lay-by is strictly worse than Save as the agent must pay the administrative fee. But it is chosen by the sophisticated quasi-hyperbolic discounter as the only feasible way to get their jacket. Without lay-by they will spend at $t = 1$ and end up with lower utility from the perspective of their $t = 0$ self.

26.1.4 Depressing the value of the bad course of action

stickK is a online platform that enables people to commit to future courses of action. stickK works as follows:

1. You state a time-based goal, such as not smoking during the next month, losing 5kg over the next 90 days, or writing the next chapter of your PhD thesis by Christmas.
2. You commit a stake that will be paid to a charity (or an anti-charity) if you fail to meet your goal.

3. At the stated time you report (or a referee appointed by you reports) whether you have met your goal.
4. If you fail to report or report that you failed to meet your goal, your credit card is debited by the staked amount.



The following is an illustration of how it works mathematically.

A sophisticated present-biased agent with $\delta = 1$ and $\beta = 1/2$ enjoys smoking. Smoking gives utility of 5. However, the agent also likes being healthy. Higher health gives utility of 8.

At $t = 0$ the agent is deciding whether to smoke over the next month ($t = 1$). If the agent doesn't smoke, it will have better health at $t = 2$.

The sophisticated agent works backward through time. At $t = 1$ its payoffs are:

$$U_1(smoking) = 5$$

$$\begin{aligned} U_1(healthy) &= \beta\delta * 8 \\ &= 4 \end{aligned}$$

The agent decides to smoke.

As a result, at $t = 0$, knowing that it will cave at $t = 1$, the agent doesn't bother committing to not smoking, even though from the perspective of $t = 0$ refraining from smoking is the better option:

$$
\begin{aligned}
U_0(smoking) &= \beta\delta * 5 \\
&= 2.5 \\
U_0(healthy) &= \beta\delta^2 * 8 \\
&= 4
\end{aligned}
$$

But now suppose the agent learns about stickK. The agent now has the option of staking a sum at $t = 0$ to prevent it from smoking. The agent decides to stake an amount equivalent to utility 5 that would be incurred at $t = 2$.

Working backward through time, the agent knows that at $t = 1$ if it has not staked any money with stickK, it will smoke. But what if it has?

$$
\begin{aligned}
U_1(stickK + smoking) &= 5 - \beta\delta * 5 \\
&= 2.5 \\
U_1(stickK + healthy) &= \beta\delta * 8 \\
&= 4
\end{aligned}
$$

The agent would refrain from smoking at $t = 1$ due to the penalty it would have to pay.

This means the agent's options at $t = 0$ are effectively:

$$
\begin{aligned}
U_0(smoking) &= \beta\delta * 5 \\
&= 2.5 \\
U_0(stickK + healthy) &= \beta\delta^2 * 8 \\
&= 4
\end{aligned}
$$

The agent chooses to commit using stickK.

For a problem of this form the agent could always successfully use stickK to commit to any action. The agent just needs to make the stake high enough.

26.1.5 Increasing the value of the optimal action: temptation bundling

"Temptation bundling" involves increasing the value of the optimal course of action by adding a temptation to that course.

Consider the following example.

Beth has the choice between the following options:

- Exercising at $t = 0$ (utility = 0) which leads her to being healthy and happy at time 1 (utility = 12)
- Watching television at time 0 (utility = 6) and being unhealthy and unhappy at time 1 (utility = 0)

Beth discounts the future quasi-hyperbolically. Beth's $\beta = 1/2$ and her $\delta = 2/3$. Does Beth exercise?

We solve as follows:

$$\begin{aligned}
U_0(exercise) &= u(x_0) + \beta\delta u(x_1) \\
&= 0 + \frac{1}{2} * 23 * 12 \\
&= 4 \\
U_0(television) &= u(x_0) + \beta\delta u(x_1) \\
&= 6 + \frac{1}{2} * 23 * 0 \\
&= 6
\end{aligned}$$

Beth chooses to watch television.

But what if Beth loved massages and remembers she has a massage voucher she has been saving? What if she decides that if she exercises today, she will go for a massage straight after? Let us assume that the utility of a massage is 3.

We solve as follows:

$$
\begin{aligned}
U_0(exercise + massage) &= u(x_0) + \beta\delta u(x_1) \\
&= 3 + \frac{1}{2} * 23 * 12 \\
&= 7 \\
U_0(television) &= u(x_0) + \beta\delta u(x_1) \\
&= 6 + \frac{1}{2} * 23 * 0 \\
&= 6
\end{aligned}
$$

Beth now chooses to exercise.

This example is not strictly a commitment device as Beth could cheat. Beth could watch television and get the massage. But people are often good at creating "mental accounts" by which they make certain money or activities out-of-bounds unless certain conditions are met.

27 Intertemporal choice applications

27.1 Savings

In the 1960s, the domestic savings rate in the Philippines was over 20% of GDP (one of the highest in Asia). By 2006, the savings rate ranges between 12 and 15 percent, far below the level of savings for most East Asian countries.

Ashraf et al. (2006) offered a commitment savings product called SEED (Save, Earn, Enjoy Deposits) to randomly chosen clients of a Philippine bank. SEED restricted access to savings for one year.

Other than providing a possible commitment savings device, no further benefit accrued to individuals with this account.

- 28.4% take-up rate of the two commitment protocols studies (either goal or time-based goal).
- Average savings balance increased by 42% after six months and by 82% after one year.

27.2 Smoking

Giné et al. (2010) tested a voluntary commitment product for smoking cessation.

Smokers were offered a product (CARES) that comprised a savings account in which they deposit funds. After six months they take urine tests for nicotine and cotinine.

If they pass, the money is returned. Otherwise, it is forfeited.

The result was that 11% of smokers offered CARES took it up. Smokers offered CARES were 3 percentage points more likely to pass the 6-month test than the control group

The effect persisted in a surprise tests at 12 months.

27.3 Gym attendance

Kirgios et al. (2020) found that:

- Teaching gym-goers how to temptation bundle with a free audiobook boosts gym visits.
- People who are given audiobooks by gyms can infer they should temptation bundle.
- Simply receiving a free audiobook with no explicit instruction boosts exercise.
- Teaching temptation bundling modestly outperforms simply giving gym-goers free audiobooks.

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